Cost, Carbon, and the Speed of Replenishment in Distribution Network Design

Abstract

A firm that distributes through regional distribution centers decides how many to operate and how fast to replenish them. Each decision trades one source of emissions against another: slower line haul burns less fuel but requires more safety stock and more stock in transit, and more distribution centers shorten the last mile but add buildings and give up the pooling that keeps inventory low. We model the two decisions jointly under a carbon price or cap and show that the design problem has a unique optimum at every carbon price. At a given network, carbon pricing favors faster replenishment when the inventory that speed saves emits more per dollar than the speed itself; because the capital tied up in goods in transit carries no carbon, financing those goods works toward slower replenishment. At a given speed, carbon pricing favors more distribution centers when a dollar of last-mile delivery emits more than a dollar of facilities and lost pooling. In the joint design, the direction of both decisions follows from four carbon-intensity-per-dollar ratios and two shares of the cost-minimizing design, and the carbon intensity of the line haul bears on the number of distribution centers only through the speed at which lost pooling is recovered. The response to a carbon price depends on the carbon charge's share of each activity's cost, which is small where cost is mostly capital and labor and larger where it is mostly fuel. A firm can compute the diagnostic from its own rate data.

Keywords: Distribution network design; Carbon pricing; Replenishment speed; Safety stock; Last-mile delivery; Continuous approximation

1 Introduction

How many distribution centers to operate and how fast to replenish them are long-standing questions in distribution network design (Eppen 1979; Daskin et al. 2002; Shen et al. 2003), and firms continue to revisit both. Some have moved stock closer to demand. Amazon divided its US fulfillment network into eight regions and reported a 19% reduction in the miles traveled to deliver packages (Amazon.com, Inc. 2023), and Home Depot is building flatbed distribution centers in its forty largest markets (The Home Depot 2020). Others have slowed their line haul. AstraZeneca has moved most of its international freight from air to ocean and ships about 65% of its cargo by sea (Pharmaceutical Commerce 2023), and truckload shippers typically convert lanes to intermodal rail once the savings reach 10 to 15% (FreightWaves 2026). The reasons these firms give are mainly cost and delivery service; Amazon also describes shorter shipments as better for the environment, and AstraZeneca relates its shift to its sustainability efforts. What is new is that carbon is beginning to be priced and reported: the European Union has priced the carbon of maritime shipping since 2024 (European Parliament and Council of the European Union 2023) and will price road fuel from 2028 (European Environment Agency 2026). California is phasing in the reporting of supply-chain emissions from 2027 (State of California 2023), and ISO 14083 standardizes how the emissions of transport and warehouses are counted (International Organization for Standardization 2023). A firm that prices or caps its distribution carbon must therefore ask which way, and how far, a carbon price moves these two decisions.

A firm serves its markets from a network of distribution centers that it replenishes over time. Among the operational choices this involves, we study two: the *density* of the network — how many distribution centers the firm operates — and the *speed* with which those centers are replenished. Both shape the firm's cost, and both shape the carbon its distribution emits. With carbon priced and reported (McKinnon and Piecyk 2012), a firm sets both with carbon as well as cost in view, and we study how a carbon price shapes that choice.

Each decision carries a standard cost–carbon trade-off. Network density governs proximity to markets: operating more distribution centers shortens the last-mile distance and changes the warehouse energy the network consumes. Replenishment speed governs availability: replenishing faster shortens the lead time that inventory must cover and changes the emissions of the freight that serves the network. Distribution emits carbon both when goods move and when they are held, and each affects both the firm's cost and its carbon. Cost balances these effects one way; with a price on carbon, the carbon-minimizing balance can fall elsewhere, and how it does so is the central question this paper addresses.

Slowing replenishment and consolidating the network are the common transport-side responses: slowing moves freight by slower, lower-emission modes, as in slow steaming in shipping (Psaraftis and Kontovas 2013; Cariou 2011), and consolidating serves markets from fewer, larger distribution centers (McKinnon and Piecyk 2012). Each response also affects inventory. A unit held in a warehouse consumes energy — refrigeration, climate control, handling — for as long as it is held. Slowing replenishment lengthens the lead time that a stock buffer must cover, so the safety stock, and the carbon of holding it, rises with the square root of that lead time; consolidating pools demand across markets (Eppen 1979) and saves facility energy while lengthening the last mile. Each substitution thus lowers one source of carbon and raises another, and whether it lowers the total depends on which change is larger.

The choice between the two sources is a choice between two operational substitutions. A firm provides *availability* either by holding safety stock or by replenishing faster — a shorter lead time leaves less for the buffer to cover, so speed and inventory are substitutes for the same service — and it provides market *proximity* either by operating more local facilities or by delivering farther from fewer, so facilities and last-mile travel are substitutes for the same access. The cost-minimizing configuration uses one mix of these activities; when carbon is priced, the optimal mix can differ. When the inventory that faster replenishment saves is more carbon-intensive per dollar than the inbound transport that would replace it, the carbon-minimizing chain, at a given network, provides availability through *faster* replenishment; a firm that finances its goods in transit saves capital as well as safety stock, and capital carries no carbon. When last-mile travel is more carbon-intensive per dollar than the facilities that would shorten it and the pooling they give up, the carbon-minimizing chain, at a given speed, provides proximity through facilities — a *more dispersed* network. The lowest-carbon distribution system thus depends on the product, on how the firm buys speed and finances its goods in transit, and on how it delivers the last mile.

We formalize this in a single-product location–inventory model with stochastic demand. The firm chooses network density and replenishment speed to minimize cost subject to a carbon cap and a service-level requirement. Cost and carbon each comprise the same four components — facilities, holding, inbound transport, and last-mile transport — and cost also includes the financing of stock in transit. The model follows the location–inventory tradition (Daskin et al. 2002; Shen et al. 2003) in a continuous-approximation form (Daganzo 2005), with a distinguishing addition: replenishment speed is a decision. Because speed sets the lead time that stochastic safety stock must cover, it directly determines inventory carbon and couples the four components together.

The paper makes three contributions. First, it characterizes the joint design. We show that at every carbon price the design problem has a unique optimum: its objective is not jointly convex in the number of distribution centers and the speed, but it is strictly convex in their logarithms. The carbon-minimizing direction of both decisions then follows from four carbon-intensity-per-dollar ratios and two shares that the cost-minimizing design reveals — the share of a distribution center's marginal cost that is lost pooling, and the share of the inventory saving from speed that is safety stock rather than stock in transit. Because faster replenishment recovers part of the pooling that dispersion gives up, the carbon intensity of line-haul freight also bears on the number of distribution centers; the link is modest, and it vanishes where the carbon-minimizing speed stays at a bound. Second, it constructs benchmark ratios from public sources and stated assumptions. Activities whose cost is mostly energy emit far more carbon per dollar than those whose cost is mostly labor or capital, so the classification depends on how speed is bought and how the last mile is driven. At the central ratios, the carbon pressure on speed bought with fuel points toward slower replenishment for every product we examine, perishables included, and that on speed bought with labor toward faster replenishment where held stock carries enough carbon, as for perishables that spoil, but toward slower replenishment for high-value goods; diesel delivery calls for more distribution centers, and electric vans for fewer when the centers are refrigerated. Third, it shows how a carbon price moves the design. The response depends on the carbon charge's share of each activity's cost, which is small at prices in current use. In a reference network that finances its stock in transit, a carbon price of \$100 per tonne lowers emissions by 5.4% at a cost increase of 0.1% through slower line haul alone, and the number of distribution centers changes only above about \$570 per tonne. A firm can compute the diagnostic from its own rate data, and the same ratios give the carbon price at which carbon becomes a material share of each activity's cost.

The remainder of the paper proceeds as follows. Section 2 positions the paper relative to the location–inventory, carbon-network-design, and inventory–transport–emissions literatures. Section 3 develops the model, Section 4 establishes the results, Section 5 applies the results to calibrated US data, Section 6 draws out the managerial implications, and Section 7 concludes.

2 Related Literature

This study connects four streams: location–inventory design with risk pooling, carbon-aware network design, the joint control of inventory and transport mode under emissions regulation, and carbon as a priced operational cost in service-constrained inventory systems. Each develops an element of the two substitutions examined here — availability (inventory versus replenishment speed) and proximity (facilities versus last-mile travel). Each of these streams is well developed, and we draw on all four; we position the contribution by what it adds to each. The paper adds the joint design of network density and replenishment speed under a carbon cap and a service requirement, an exact classification of the direction in which carbon moves both decisions, and a calibration of the carbon intensities that decide it.

2.1 Location–inventory design and risk pooling

A first stream establishes how demand risk pooling shapes distribution network design: jointly choosing facility location and safety stock, it characterizes the cost-optimal number and placement of distribution centers. Eppen (1979) characterizes the effect of centralization on expected cost in the multi-location newsvendor, establishing the square-root relationship between the number of stocking locations and pooled safety stock. Daskin et al. (2002) and Shen et al. (2003) embed working and safety stock into the facility-location decision, so that risk pooling shapes the optimal number and placement of distribution centers, with extensions to stochastic settings by Snyder et al. (2007), to capacitated warehouses by Özsen et al. (2008), and to continuous-approximation allocation by Tsao et al. (2012). Daganzo (2005) provides the continuous-approximation foundation in which last-mile distance scales with the inverse square root of the number of facilities. Closest to our spatial–inventory coupling, Straubert (2024) ties the number of stocking locations to an exact reorder-point/order-quantity inventory model through a delivery-lead-time penalty, solved by continuous approximation. Recent work extends the location–inventory problem in directions close to ours. Hu et al. (2025) add the choice of storage temperature, trading cooling cost against the quality decay of fresh products in a network with risk pooling, and Jalal et al. (2026) integrate multi-period location, inventory, and transportation decisions under demand uncertainty for a pharmaceutical network. Puttemans et al. (2025) review this literature and call for more realistic modeling assumptions; our calibration in Section 5 anchors each carbon rate in public data. The centralize-versus-decentralize question these models pose can also depend on forces beyond demand pooling: Schmitt et al. (2015) show analytically that supply-disruption risk diversification can favor decentralization for a risk-averse firm, in contrast to the classical pooling result. These works develop network density and safety-stock placement under risk pooling, with a cost or service objective and no carbon; they do not make replenishment speed a continuous decision coupled with carbon. The present study retains the risk-pooling structure and the density decision and adds an endogenous replenishment speed together with a carbon objective.

2.2 Carbon-aware network design

A second stream brings carbon into facility location and network design, characterizing how an emissions objective or cap can change the optimal network configuration. Cachon (2014) relates retail store density to greenhouse gas cost, with consumer travel to stores as the dominant emission channel and inventory entering through retail floor space. Turken et al. (2017) examine facility location and capacity under a carbon tax and emission limits, weighing centralization against decentralization through production scale economies; Harris et al. (2011) compare cost-optimal and emissions-aware network designs numerically; Wang et al. (2020) develop a carbon-compliant location–inventory model under demand and carbon-price uncertainty; Li et al. (2017) jointly optimize distribution-network design and two-echelon inventory under a CO_2 emission tax; and Hacardiaux and Tancrez (2020) assess the transport-emission savings of a shared, consolidated location–inventory network under horizontal cooperation, with Hacardiaux et al. (2022) balancing partner preferences over logistics cost and carbon footprint in the same cooperative setting. Closest to ours in method, Zhang et al. (2026) combine continuous approximation with integer programming to show how a carbon price reshapes the choice among drone, truck–drone, and truck last-mile delivery. These studies make density or facility configuration the central decision and bring emissions into the objective, the constraints, or an ex post assessment; where transport is modeled as an operational choice, it enters through travel distances, discrete modes, or lead-time parameters rather than through a continuously optimized replenishment speed chosen jointly with density. The present study treats network density and replenishment speed as joint decisions and resolves the proximity substitution between facilities and last-mile travel analytically.

2.3 Inventory, transport mode, and emissions

A third stream couples transport-mode and lead-time choices with inventory control under emissions regulation, characterizing how carbon reshapes the trade-off between freight and the inventory it serves. Hoen et al. (2014) and Chen and Wang (2016) study transport mode selection alongside inventory under stochastic demand and emissions regulation; Qi et al. (2022) select door-to-door transport services and routes for China–Europe freight against transport cost, in-transit inventory cost, and carbon; Bektaş and Laporte (2011) formalize the load–emission relationship in the pollution-routing problem, embedding fuel use and emissions as a function of vehicle load in the routing objective; Moradi et al. (2025) optimize driving speed and path selection under load- and grade-dependent fuel use at the routing level; Konur and Schaefer (2014) solve the economic order quantity model with less-than-truckload and truckload carriers under carbon cap, tax, and cap-and-trade regulation, and Schaefer and Konur (2015) extend this to a continuous-review, reorder-point inventory system under stochastic demand, characterizing the cost–emission Pareto frontier and analyzing lead-time and demand-variance sensitivity. Drent et al. (2023) analyze dynamic dual supply-mode selection under an emission target, and Hammami et al. (2015) bring carbon from manufacturing, ordering, and inventory into a multi-echelon production–inventory model with lead-time constraints. In shipping, Maloni et al. (2013) show by simulation that slower vessel speed lowers both total cost and carbon on an Asia–North America lane while raising the pipeline inventory that shippers carry. Our speed threshold prices the carbon as well as the cost of the inventory that slower replenishment requires. Across these papers, transport service is represented through route/load physics, carrier class, discrete supply mode, or lead-time parameters and constraints; replenishment speed is not a continuous decision chosen jointly with network density. The present study adds to this stream the continuous speed–density threshold that results when replenishment speed and network density are chosen jointly under a carbon cap and a service requirement.

Boone and Ganeshan (2025) choose a discrete freight mode against inventory under an emission cap at a single location, with no network-density decision. The present study replaces the finite mode set with a continuous replenishment-speed decision, prices carbon on the holding as well as the transport side, embeds it in a network whose density is chosen jointly, and characterizes the resulting continuous speed–density threshold and complementarity.

2.4 Carbon policy, service, and safety-stock placement

A fourth stream prices carbon directly into inventory and service decisions, showing how a carbon price or cap shifts order quantities and safety-stock placement. Hua et al. (2011) and Bouchery et al. (2012) develop carbon-aware order-quantity models under cap-and-trade and multi-objective formulations, and Benjaafar et al. (2013) position cap, tax, and cap-and-trade as priced operational instruments in simple supply-chain models; Chen et al. (2013) derive the closed-form carbon-constrained economic order quantity, the single-location analog of the marginal carbon-intensity logic we extend to a network, and Arıkan and Jammernegg (2014) impose a product carbon-footprint constraint on a dual-sourcing newsvendor. Silbermayr et al. (2017) bring environmental constraints into inventory pooling with dependent demand. Within multi-echelon service models, Ni and Shu (2015) study the trade-off between service time and carbon in safety-stock placement under the guaranteed-service framework, and Wang et al. (2025) incorporate cap-and-trade into guaranteed-service placement. Probabilistic carbon limits have been modeled through chance constraints in lot sizing (Koca 2021) and in supply-chain network design (Shaw et al. 2016), typically inside large mixed-integer programs; we instead obtain a closed-form deterministic equivalent that scales the transport component by a single factor. A recent review (Sebatjane 2025) surveys carbon-aware inventory models and the regulatory instruments they treat; across this body of work, transport-emission modeling, service and inventory, network design, and demand or carbon uncertainty have developed in largely separate streams, and we bring these strands together. The service-placement models in particular price carbon against service and safety stock, with the joint design of network density and replenishment speed outside their scope.

Prior work covers carbon-aware density, mode and lead-time choice, service-constrained placement, and chance constraints on carbon. The present study couples endogenous replenishment speed with network density under carbon and service constraints and shows that the joint design is unique at every carbon price. It classifies the carbon-minimizing direction of both decisions exactly. A speed threshold compares the carbon intensity per dollar of the inventory saving from speed — holding, weighted by the safety-stock share of that saving — with that of the speed it buys; a density threshold compares last-mile intensity with a pooling-weighted blend of facility, holding, and line-haul intensities; and complementarity links the two. It then estimates these intensities from public data.

3 The Model

3.1 Operational environment

Region and markets. A firm serves N markets distributed over a service region of area A. Per-period demand at market i is a random variable with mean μ and standard deviation σ, identically distributed across markets with common pairwise correlation $\rho \in [0,1)$. Demand is stationary and the setting is single-product. Markets and distribution centers are treated symmetrically, the standard simplification that admits closed-form continuous-approximation results (Daganzo 2005).

Distribution centers and geography. The firm opens n identical distribution centers (DCs), $1 \le n \le N$, each serving a sub-region of area A/n. By the continuous-approximation spatial law, the mean last-mile (DC-to-market) distance falls as the square root of the served area,

$$d(n) \;=\; \delta\, n^{-1/2}, \qquad \delta \propto \sqrt{A}, \tag{1}$$

so a denser network shortens delivery but only at a decreasing rate (Daganzo 2005).

Replenishment and the speed decision. Each DC is replenished directly from the source at a chosen speed $v \in [v_{\min}, v_{\max}]$. Individual DCs lie at different distances from the source; what enters cost and carbon is total inbound freight: with identical DCs each carrying throughput $N\mu/n$, this is $\sum_j (N\mu/n)\, D_j = N\mu\, \bar{D}$, the total throughput times the *mean* source-to-DC distance $\bar{D}$. Because the DCs fill a region of fixed size served from a fixed source, $\bar{D}$ is set by the region's geometry (order $\sqrt{A}$) and is approximately invariant to n; we set $D \equiv \bar{D}$ and treat it as constant.[1] The replenishment lead time is then

$$L(v) \;=\; D/v. \tag{2}$$

Faster replenishment is simultaneously more expensive and more carbon-intensive. The maintained assumption is only that the inbound cost rate $t(v)$ and emission rate $\varepsilon(v)$ (both per unit per mile) are increasing and convex in v; for the closed-form results, we use the quadratic instantiation $\varepsilon(v) = \varepsilon_0 v^2$ and $t(v) = t_0 + t_1 v^2$, the canonical form motivated by the cubic fuel-speed law of slow steaming (Psaraftis and Kontovas 2013; Cariou 2011). We interpret v as a general index of replenishment speed on a given origin–destination lane. Services on the lane — intermodal rail, truckload with a solo driver, or truckload with a team of two drivers who keep the truck moving about twice as many hours a day — differ in the lead time they give, and slow steaming is one motivating case. Treating replenishment speed as a continuously optimized decision, chosen jointly with network density, couples the transport and facility-and-holding components in this distribution setting.

Fuel and other costs of replenishment. Services differ in how they shorten lead time. The transport cost rate divides into fuel and other costs, $t(v) = t^{\mathrm{f}}(v) + t^{\mathrm{o}}(v)$. With a fixed fuel price and emission factor, and with fuel accounting for the emissions that vary with speed, $\varepsilon'(v) = \chi_{\mathrm{f}}\, t^{\mathrm{f}\prime}(v)$, where χ_{f} is the carbon per dollar of fuel, so that

$$\frac{\varepsilon'(v)}{t'(v)} = \chi_{\mathrm{f}}\, \xi(v), \qquad \xi(v) = \frac{t^{\mathrm{f}\prime}(v)}{t'(v)}. \tag{3}$$

When both cost components rise with speed, $\xi(v)$ is the fuel share of the marginal cost of speed. Under the quadratic instantiation, $t_1 = b_{\mathrm{f}} + b_{\mathrm{o}}$ and $\varepsilon_0 = \chi_{\mathrm{f}} b_{\mathrm{f}}$, where $b_{\mathrm{f}} v^2$ and $b_{\mathrm{o}} v^2$ are the speed-dependent fuel and other costs, so that ε_0/t_1 is constant. Moving from intermodal rail to truckload raises fuel use per unit-mile, whereas moving from a solo driver to a team raises driver cost at nearly the same fuel use per mile. Where t is a freight rate, the relation assumes that fuel costs pass through to the rate.

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[1] Two modeling reductions are embedded here. First, the symmetric-DC reduction assigns every DC the representative lead time $L = \bar{D}/v$ in place of its own $L_j = D_j/v$. Safety stock enters through $\sqrt{L}$, which is concave in distance, so by Jensen's inequality $\sqrt{\bar{D}} \geq n^{-1} \sum_j \sqrt{D_j}$ (Jensen 1906): using the mean distance slightly *overstates* aggregate safety stock, by a gap that is second-order in the dispersion of the D_j and reduces as the distances equalize. Second, replenishment is point-to-point; were the DCs served on a consolidated route, the inbound distance would grow as $\sqrt{n}$ in the manner of the store-delivery tour of Cachon (2014), adding a term that reinforces centralization in Proposition 5 and leaving the fixed-density speed comparison of Proposition 3 unchanged; with inbound distance varying in n, however, the speed response would depend on density through $D(n)$, so the complementarity of Proposition 6 need not carry over unchanged.

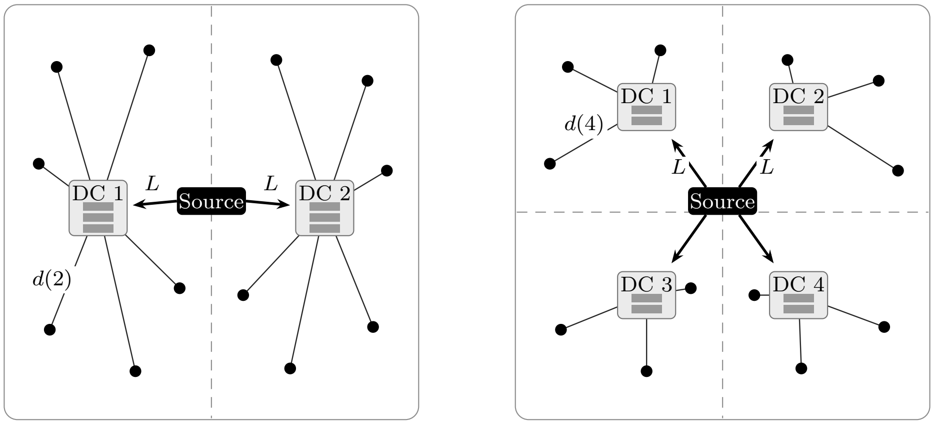
Inventory and service. Each DC operates a base-stock policy and must meet a cycle service level α — the in-stock probability at the end of a period, the Type-1 (cycle) service measure. With normally distributed demand the safety factor is $z = \Phi^{-1}(\alpha)$, with $\alpha > \frac{1}{2}$ so that $z > 0$, taken here as a service requirement (a parameter). Demand is independent across periods, and each DC reviews its inventory continuously, so safety stock protects against demand over the replenishment lead time; under periodic review the protection interval would also include the review period. A DC that pools $g = N/n$ markets faces aggregate per-period demand with standard deviation $\sigma\sqrt{g\,[1 + (g-1)\rho]}$, the statistical economy of scale in inventory (Eppen 1979). Covering lead-time demand, each DC holds safety stock $z\,\sigma\sqrt{L}\,\sqrt{g[1 + (g-1)\rho]}$. Summing over the n DCs and defining the *pooling expression*

$$\Psi(n) \;=\; N\rho + n(1-\rho), \tag{4}$$

the system-wide safety stock is

$$\mathrm{SS}(n, v) \;=\; z\,\sigma\,\sqrt{D/v}\,\sqrt{N\,\Psi(n)}. \tag{5}$$

At $\rho = 0$, (5) reduces to $z\sigma\sqrt{L}\sqrt{nN}$, the square-root law: decentralizing from 1 to N DCs multiplies safety stock by $\sqrt{N}$. As $\rho \to 1$, $\Psi(n) \to N$ and safety stock becomes invariant to n — the pooling benefit vanishes.



$n = 2$: each DC serves 6 markets
more stock per DC, longer delivery legs

$n = 4$: each DC serves 3 markets
less stock per DC, shorter delivery legs

Faster replenishment (shorter L), at a given density:
system-wide safety stock shrinks, so holding cost and carbon fall, while freight cost and carbon rise.

More DCs (higher n), at a given speed:
delivery legs $d(n)$ shorten with territory area, so system-wide last-mile cost and carbon fall, while facility cost and carbon rise and weaker pooling raises system-wide holding cost and carbon.

Figure 1: Model representation for 12 markets (N) and 2 or 4 DCs ($n = 2$ or 4).

Table 1: Notation. The decision variables are the network density n and the replenishment speed v; all other quantities are exogenous parameters or derived.

Symbol	**Meaning**
Decision variables	
n	number of DCs (network density), $1 \leq n \leq N$
v	replenishment speed, $v \in [v_{\min}, v_{\max}]$
Demand (exogenous)	
N	number of markets
$\mu,\ \sigma$	mean, standard deviation of demand per market per period
ρ	cross-market demand correlation, $\rho \in [0, 1)$
Geography, cost, carbon (exogenous)	
$A,\ D,\ \delta$	region area; inbound distance; last-mile distance constant
$F,\ \varphi$	fixed cost, carbon per DC per period
$h,\ \eta$	holding cost, carbon per unit per period
h_p	financing cost of stock in transit per unit per period
$t(v),\ \varepsilon(v)$	inbound transport cost, carbon per unit-mile (convex in v)
$t_L,\ \varepsilon_L$	last-mile transport cost, carbon per unit-mile
α	cycle service level (in-stock prob.); $z = \Phi^{-1}(\alpha)$
$\tau,\ B$	carbon price; carbon cap
Derived	
$g = N/n$	markets served per DC
$L = D/v$	replenishment lead time
$d(n) = \delta\, n^{-1/2}$	mean last-mile distance
$\Psi(n) = N\rho + n(1-\rho)$	pooling expression
$\mathrm{SS}(n, v)$	system-wide safety stock, eq. (5)
$P(v) = N\mu D/v$	stock in transit
$a(n, v)$	safety-stock share of the inventory saving from speed, eq. (8)

Decisions. The firm chooses the network density $n \in [1, N]$ (relaxed to a continuous variable and rounded) and the replenishment speed $v \in [v_{\min}, v_{\max}]$. Service is a requirement, so α (hence z) is a parameter in our model. Table 1 collects the notation.

3.2 Cost and carbon

Both the cost and carbon functions are system-wide flows over a single representative period, consistent with the stationary single-product environment above: F, φ are charged per DC per period, h, η per unit held per period, and the transport terms accrue against the per-period throughput. The representative distances in Figure 1 therefore enter global cost and carbon through aggregate freight tasks, not through a per-territory objective. Throughput itself is conserved at $N\mu$ regardless of the configuration: the firm moves the mean demand of all N markets through the network, whatever density n and speed v it chooses. With per-period throughput $N\mu$,

$$K(n,v) = \underbrace{F\,n}_{\text{fixed}} + \underbrace{h\,\mathrm{SS}(n,v)}_{\text{holding}} + \underbrace{h_p\,P(v)}_{\text{in transit}} + \underbrace{t(v)\,N\mu D}_{\text{inbound}} + \underbrace{t_L\,N\mu\,d(n)}_{\text{last-mile}}, \tag{6}$$

$$C(n,v) = \underbrace{\varphi\,n}_{\text{fixed}} + \underbrace{\eta\,\mathrm{SS}(n,v)}_{\text{holding}} + \underbrace{\varepsilon(v)\,N\mu D}_{\text{inbound}} + \underbrace{\varepsilon_L\,N\mu\,d(n)}_{\text{last-mile}}, \tag{7}$$

where F, φ are fixed cost/carbon per DC, h, η are holding cost/carbon per unit per period, h_p is the financing cost of a unit in transit per period, $P(v) = N\mu D/v$ is the stock in transit, and t_L, ε_L are last-mile cost/carbon rates (Shen et al. 2003; Hua et al. 2011; Benjaafar et al. 2013). Each term multiplies a rate by the activity it prices; apart from the financing of stock in transit, which carries no carbon, the two functions differ only in whether the rate is measured in dollars or kilograms.

Facilities. The facility terms Fn and φn are the periodic cost and carbon of operating the open DCs. They rise linearly in n and do not depend on v. Facility carbon φ is distinct from holding carbon η: φ captures baseline building energy and embodied carbon, whereas η prices the incremental carbon of storing a unit (for example refrigeration or handling).

Holding. The holding terms $h\mathrm{SS}(n,v)$ and $\eta\mathrm{SS}(n,v)$ value the system-wide safety stock of (5). They rise in n through the loss of pooling in $\Psi(n)$ and fall in v through the lead-time factor $\sqrt{D/v}$. This is the channel through which replenishment speed couples transport and inventory carbon; for cost, stock in transit is a second channel.

Stock in transit. Goods in transit, $P(v) = N\mu D/v$, fall as replenishment speeds up. A firm that owns them while they travel carries their financing cost, h_p per unit per period — the capital charge on the value of the goods — and $h_p = 0$ when title passes on receipt at the distribution center. Stock in transit occupies no warehouse, and the carbon of moving it is counted in the inbound transport term, so it adds no carbon of its own. Its financing cost does not depend on n. Because that cost carries no carbon, it lowers the carbon intensity of the inventory saving that speed obtains: at a configuration (n,v), the share of that saving that is safety stock is

$$a(n,v) = \frac{h\,\mathrm{SS}(n,v)}{h\,\mathrm{SS}(n,v) + 2h_p\,P(v)} \in (0,1], \tag{8}$$

where the factor two reflects that safety stock falls with the square root of the lead time and stock in transit with the lead time itself. Without financing, $a = 1$; the share rises with speed.

Inbound transport. The inbound terms multiply the representative freight task $N\mu D$ by the speed-dependent rates $t(v)$ and $\varepsilon(v)$. Since D is treated as invariant to n, inbound transport does not vary with network density; it rises in v through the increasing convex rates $t(v)$ and $\varepsilon(v)$, opposing the holding term that falls in v.

Last-mile transport. The last-mile terms multiply throughput $N\mu$ by the mean DC-to-market distance $d(n) = \delta n^{-1/2}$ and the rates t_L, ε_L. A denser network shortens this leg at the continuous-approximation rate, so the last-mile term is decreasing and convex in n and independent of v under constant per-mile last-mile rates.

Decomposition and scope. The carbon function splits into a transport component $\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)$ and a facility-and-holding component $\varphi n + \eta \mathrm{SS}(n, v)$; the cost function has the analogous partition, with the financing term $h_p P(v)$ in addition. Slowing replenishment reduces inbound transport carbon but raises safety stock, while dispersion reduces last-mile distance but raises facility count and pooling loss. Two scope points follow. Cycle stock depends on the replenishment quantity, which we hold fixed, and is invariant to both decisions. Production and procurement are excluded because throughput is fixed at $N\mu$ and therefore constant in both decisions.

3.3 The program

The firm minimizes cost subject to a carbon cap B and the service requirement,

$$\min_{1\le n\le N,\ v_{\min}\le v\le v_{\max}} K(n, v) \quad \text{s.t.} \quad C(n, v) \le B. \tag{9}$$

Here K and C are expected flows over the model's accounting period: under the stationary demand assumption they are the long-run average cost and carbon of a configuration, and they can be scaled to an annual expected-carbon cap by multiplying by the number of periods. Because demand is random, realized carbon over a specified accounting horizon fluctuates about its expectation; we develop the analysis for the expected-value cap and return below to a chance-constrained cap that bounds realized carbon with specified confidence (Proposition 10).

Equivalently, pricing carbon at $\tau \ge 0$ gives the priced objective $J = K + \tau C$. At an interior optimum, the cap multiplier equals the price ($\lambda = \tau$), as in carbon-constrained inventory models (Chen et al. 2013); sweeping τ traces the cost–carbon frontier, along which the optimal configuration moves continuously (Proposition 2). Because cost and carbon are convex in the logarithms of n and v (Proposition 2), the capped problem (9) is a convex program in those variables: every cap above the carbon minimum is met by a finite carbon price, and the capped problem and the problem under that price share the same unique optimum. The carbon minimum itself is the limit as the price grows without bound. Jumps arise only from the integer number of distribution centers. Define the blended rates

$$F_\tau = F + \tau\varphi, \qquad H_\tau = h + \tau\eta, \qquad T_\tau(v) = t(v) + \tau\varepsilon(v), \qquad \ell_\tau = t_L + \tau\varepsilon_L.$$

Then

$$J(n, v; \tau) \;=\; F_\tau n + H_\tau\,\mathrm{SS}(n, v) + h_p\,P(v) + T_\tau(v)N\mu D + \ell_\tau N\mu d(n). \tag{10}$$

The financing cost of stock in transit carries no carbon and is not blended. Setting $\tau = 0$ recovers cost; the limit $\tau \to \infty$ recovers carbon up to scale.

4 Analysis

We now characterize how carbon pricing changes the joint design. We first show that the design problem has a unique optimum at every carbon price (Proposition 2). The first-order conditions then decompose into two marginal substitutions, each analyzed with the other decision held fixed: Proposition 3 gives the speed threshold on the availability margin, Proposition 4 records the service and volatility comparative static, and Proposition 5 gives the pooling-adjusted density threshold on the proximity margin. Proposition 6 establishes the complementarity through which the two margins interact, and Proposition 7 combines them into an exact classification of the carbon-minimizing direction of both decisions. Corollary 8 gives the magnitudes in terms of the four ratios and the share a_K when pooling is absent, in closed form without financing, and Proposition 9 describes how the design moves as the carbon price rises. Section 5 applies them to a calibrated network. Proofs are in Appendix A.

Lemma 1 (Carbon intensity per dollar)**.** *Pricing carbon at τ replaces each cost rate in* (6) *by the blended rate* $\mathrm{rate}_\$ + \tau\, \mathrm{rate}_{\mathrm{kg}}$*, yielding the objective J of* (10)*; the financing cost of stock in transit is not blended. In the interior comparisons below, each holding the other decision fixed, the* direction *in which the carbon-minimizing choice departs from the cost-minimizing one is governed by four carbon-intensity-per-dollar ratios — holding η/h, inbound transport at the margin ε'/t' of* (3) *(ε_0/t_1 under the quadratic instantiation), last-mile ε_L/t_L, and facilities φ/F — and, on the speed margin, by the safety-stock share a of* (8)*, while the magnitude and the realized joint optimum also depend on pooling, the bounds, and the complementarity of the two decisions.*

The carbon-minimizing choice on each decision therefore depends on each activity's carbon intensity *relative to its cost*, not on its absolute rates.

With $G(n, v) = \partial \mathrm{SS}(n, v)/\partial n$ denoting the marginal safety stock of an additional DC, differentiation gives the stationarity conditions

$$F_\tau + H_\tau G(n, v) = \frac{\ell_\tau N \mu \delta}{2}\, n^{-3/2}, \tag{11}$$

$$N \mu D\, T'_\tau(v) = \frac{H_\tau\, \mathrm{SS}(n, v)}{2v} + \frac{h_p\, P(v)}{v}, \tag{12}$$

the first balancing the marginal facility and lost-pooling costs against the marginal last-mile saving, and the second balancing the marginal transport cost of speed against the marginal holding and financing saving it obtains. The joint first-order conditions thus reveal two marginal substitutions. The speed condition (12) governs the *availability margin*: safety stock and replenishment speed are alternative ways to provide service. The density condition (11) governs the *proximity margin*: facilities and last-mile distance are alternative ways to reach markets. The thresholds below compare the carbon intensity per dollar of the activities on each margin, and complementarity determines how the two margins interact in the joint configuration.

The stationarity conditions identify the optimum because the design problem has a unique optimum at every carbon price. The objective is not jointly convex in n and v, but it is convex in their logarithms. Let $J^{\mathrm{r}}(n;\tau) = \min_{v_{\min} \le v \le v_{\max}} J(n, v; \tau)$ be the *reduced objective*, the cost, carbon charge included, of operating n distribution centers when speed is chosen optimally for that network. A function on an interval is *strictly quasiconvex* if it strictly decreases and then strictly increases, either phase possibly empty, so that it has a single minimizer.

Proposition 2 (A unique optimum)**.** *Let t and ε be increasing, differentiable, and convex, let all cost and carbon rates be positive, let $h_p \ge 0$, and let $v_{\min} > 0$. For every carbon price $\tau \ge 0$, and for the carbon objective C:* (i) *the objective is strictly convex in* $(\ln n, \ln v)$ *on the feasible region;* (ii) *the problem has a unique minimizer, and under a carbon price the minimizer $\big(n^*(\tau), v^*(\tau)\big)$ is continuous in τ;* (iii) *the speed response $v^*(n)$ is unique, continuous, and nondecreasing in n, and the objective at any fixed speed and the reduced objective are strictly quasiconvex in n on* $[1, N]$.

Proposition 2 has a direct interpretation. Adding a distribution center shortens every delivery but adds a facility charge and the safety stock lost to pooling. Measured in proportional changes of the network and of speed, each of these effects is convex, so once an additional center stops paying for itself, no larger network makes it pay again. Three consequences follow. First, the stationarity conditions (11)–(12) identify the optimum whenever it is interior. Second, the comparative statics below are global, not local to a particular optimum. Third, in the continuous model the design moves without jumps as the carbon price rises, so discontinuities arise only from the integer number of distribution centers. Because the reduced objective is flat near its minimum, treating that number as continuous costs little. In the calibrated network of Section 5, rounding it to the best integer raises the objective by at most 0.09%, in line with the accuracy reported for continuous-approximation designs (Ouyang and Daganzo 2006; Ansari et al. 2018).

Availability: inventory or speed. The speed condition isolates the availability margin. Increasing speed raises inbound transport cost and carbon but lowers the safety stock required for service; the speed threshold identifies which activity is more carbon-intensive per dollar on this margin.

Proposition 3 (Speed reversal)**.** *Fix the density n, and let the speed-dependent cost and carbon be*

$$\hat{K}(v) = h\,\kappa(n)\,v^{-1/2} + h_p N\mu D\,v^{-1} + N\mu D\,t(v), \qquad \hat{C}(v) = \eta\,\kappa(n)\,v^{-1/2} + N\mu D\,\varepsilon(v),$$

with $\kappa(n) = z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}$. If t and ε are increasing, differentiable, and convex, then at a common density and for interior optima the carbon-minimizing speed exceeds the cost-minimizing speed if and only if the inventory saving from speed is more carbon-intensive per dollar than inbound transport at the margin*,*

$$a(n, v_{\mathrm{K}})\,\frac{\eta}{h} \;>\; \frac{\varepsilon'(v_{\mathrm{K}})}{t'(v_{\mathrm{K}})}, \tag{13}$$

evaluated at the cost-minimizing speed v_{K}*. With the quadratic instantiation* $\varepsilon(v) = \varepsilon_0 v^2$*,* $t(v) = t_0 + t_1 v^2$*, the right side is* ε_0/t_1*. Without financing (*$h_p = 0$*),* $a = 1$*, the threshold reduces to* $\eta/h > \varepsilon_0/t_1$*, and the unconstrained optimum is closed form,*

$$v^*(n) \;=\; \left[\frac{(h+\tau\eta)\, z\sigma\sqrt{ND}\,\sqrt{\Psi(n)}}{4\,(t_1+\tau\varepsilon_0)\,N\mu D}\right]^{2/5}. \tag{14}$$

On $[v_{\min}, v_{\max}]$ *the optima are the projections of the unconstrained speeds onto the interval; under the threshold* (13) *the comparison* $v_C \geq v_K$ *then holds weakly, and strictly when neither projection binds.*

Financing stock in transit makes speed more valuable to the cost-minimizing firm but not to the carbon-minimizing one, whose objective it does not enter; it lowers the share a and so works against faster replenishment under carbon. A lower fuel share of the marginal cost of speed, by (3), works the other way. Under the quadratic instantiation, as the carbon price rises, the fixed-density speed optimum moves monotonically between the cost and carbon endpoints, with or without financing (Proposition 9), so the endpoint comparison of the proposition also orders every finite price.
Total carbon is U-shaped in speed. When the inventory saving from speed is more carbon-intensive per dollar than the freight that would replace it, a tighter carbon target can select *faster* replenishment. Volatility and the service target raise the level of speed at both optima. Under the quadratic instantiation, without financing they leave its direction unchanged, and with financing they raise the share a and so can turn the comparison toward faster replenishment (Proposition 4). The comparison is at common density; in the joint problem, the density channel reinforces the speed reversal when carbon also disperses the network and attenuates it otherwise, and Proposition 7 separates the two effects exactly.

Service and volatility enter through the same channel. We use $J^{\mathrm{r}}(n;s) \equiv J\big(n, v^*(n;s); s\big)$, with $s = z\sigma$, to denote the reduced objective obtained after substituting the speed response into the priced objective.

Proposition 4 (Service and volatility as carbon parameters)**.** *Under the quadratic instantiation,* $v^*(n)$ *is increasing in the safety factor* $z = \Phi^{-1}(\alpha)$ *and in the demand standard deviation* σ *wherever it is interior, and at every* n *both raise the holding term in* (11)*. Consequently, at every carbon price, a higher* z *or* σ *raises the inventory penalty of a denser network and lowers the optimal density, strictly when it is interior; in particular, it shifts the carbon-minimizing density toward consolidation. With financing (*$h_p > 0$*), a higher* z *or* σ *also raises the share* $a(n, v_{\mathrm{K}})$*, so at a common density it can turn the comparison* (13) *toward faster replenishment but never toward slower.*

A service requirement therefore affects the carbon-minimizing design as well as inventory cost; endogenous service choice is a natural extension (Section 7).

Proximity: facilities or last-mile travel. The density condition isolates the proximity margin. Increasing density shortens the last mile but raises facility cost and carbon and weakens pooling; the density threshold identifies when carbon places more weight on last-mile distance than on the facility-and-pooling cost of proximity, and pooling keeps the comparison from reducing to a single pair of rates.

Proposition 5 (Density and dispersion)**.** *Fix the speed v and let*

$$G(n,v) \;=\; \frac{\partial\,\mathrm{SS}(n,v)}{\partial n} \;=\; z\sigma\sqrt{D/v}\,\sqrt{N}\,\frac{1-\rho}{2\sqrt{\Psi(n)}} \;\geq\; 0 \tag{15}$$

be the marginal safety-stock penalty of an additional DC. The minimizer of $J(\cdot,v)$ is unique; it is interior if and only if $J_n(1,v) < 0 < J_n(N,v)$, and otherwise lies at a boundary, $n^* \in \{1,N\}$. For an interior cost-minimizing density $n_K(v)$, the carbon-minimizing density exceeds it, $n_C(v) > n_K(v)$, if and only if last-mile transport is more carbon-intensive per dollar than the pooling-adjusted facility threshold*,*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\, G\big(n_K(v),v\big)}{F + h\, G\big(n_K(v),v\big)}. \tag{16}$$

The right-hand side is a weighted average of the facility ratio φ/F and the holding ratio η/h, with weight on η/h rising in the pooling penalty G: it equals φ/F when pooling is absent and tilts monotonically toward η/h as G grows (lower correlation, higher safety factor or volatility, or slower replenishment). In the no-pooling limit ($\rho \to 1$, so $G \to 0$) the optimum is closed form when interior, $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ (with the blended last-mile and facility rates $\ell_\tau = t_L + \tau\varepsilon_L$ and $F_\tau = F + \tau\varphi$ of (10)*), and* (16) *reduces to*

$$\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi}{F}. \tag{17}$$

The density decision is the centralize-versus-decentralize question in carbon terms. By (16), the carbon-minimizing network is more dispersed when the last-mile ratio ε_L/t_L exceeds a facility threshold that pooling shifts from φ/F toward η/h. Pooling therefore enters as an explicit weight on the holding ratio, not as a uniform centralizing force: when holding is more carbon-intensive per dollar than facilities, stronger pooling raises the threshold for dispersion; otherwise it lowers it. A corner optimum, when it arises, is fully centralized or fully decentralized.

Complementarity of speed and density. The two margins interact through safety stock: faster replenishment reduces the inventory penalty of density, and higher density increases the inventory exposure affected by speed. Proposition 6 makes the interaction precise.

Proposition 6 (Complementarity)**.** *For $\rho < 1$, the speed response $v^*(n)$ is nondecreasing in n and the density response $n^*(v)$ is nondecreasing in v, each strictly where it is interior.*

Faster replenishment shrinks safety stock and lowers the inventory penalty of dispersion; a denser network raises the inventory at stake and raises the value of speed. Speed investments and network density are therefore best evaluated jointly.

The joint design. The marginal thresholds hold one decision fixed. In the joint design both decisions adjust, and complementarity links them: a carbon price that disperses the network also raises the value of speed, and one that speeds replenishment lowers the inventory penalty of dispersion. Proposition 7 resolves the joint comparison exactly. We use $r_h = \eta/h$, $r_t = \varepsilon_0/t_1$, $r_L = \varepsilon_L/t_L$, and $r_F = \varphi/F$ to denote the four ratios of Lemma 1, so that the two diagnostic ratios are

$$R_{\text{speed}} = \frac{r_h}{r_t}, \qquad R_{\text{density}} = \frac{r_L}{r_F},$$

and let (n_K, v_K) and (n_C, v_C) denote the cost- and carbon-minimizing configurations, $\hat{v}_C$ the carbon-minimizing speed at the cost-minimizing density n_K, and $a_K = a(n_K, v_K)$ the safety-stock share at the cost minimum. Two quantities of the cost-minimizing network enter. The first is the *pooling weight*

$$\omega = \frac{h\,G(n_K, v_K)}{F + h\,G(n_K, v_K)} = 1 - \Big(\frac{n_K}{n_K^0}\Big)^{3/2}, \qquad n_K^0 = \Big(\frac{t_L N \mu \delta}{2F}\Big)^{2/3}, \tag{18}$$

the share of the marginal cost of a distribution center that is lost pooling rather than facility cost. Since n_K^0 is the unconstrained density the firm would choose if pooling were absent, ω measures how far pooling holds the network below that size. The second, for interior n_K, is the pair of *speed bands*

$$\underline{R} = \sqrt{\Psi(n_K)/N} \;<\; 1 \;<\; \overline{R} = \sqrt{\Psi(n_K)/\Psi(1)}, \tag{19}$$

which bound how far a change in density can move the carbon-minimizing speed.

Proposition 7 (Joint carbon-minimizing design)**.** *Suppose that $\rho < 1$ and that the cost-minimizing density n_K is interior.*

(i) Density. *The carbon-minimizing network is more dispersed than the cost-minimizing one, $n_C > n_K$, if and only if*

$$r_L \;>\; (1-\omega)\, r_F \;+\; \omega\, r_h \sqrt{v_K/\hat{v}_C}, \tag{20}$$

with $n_C = n_K$ at equality and $n_C < n_K$ when the inequality is reversed. Under the quadratic instantiation, and when v_K and $\hat{v}_C$ are interior, (20) *becomes*

$$r_L \;>\; (1-\omega)\, r_F \;+\; \omega\, r_h^{4/5}\, r_t^{1/5}\, a_K^{-1/5}. \tag{21}$$

(ii) Speed. *Under the quadratic instantiation, and when the speed responses of the cost and the carbon objective are interior at n_K and at n_C, the carbon-minimizing speed satisfies*

$$\ln \frac{v_C}{v_K} = \frac{2}{5} \ln \big(a_K R_{\text{speed}}\big) + \frac{1}{5} \ln \frac{\Psi(n_C)}{\Psi(n_K)}, \tag{22}$$

so that $v_C > v_K$ *if and only if* $a_K R_{\text{speed}} > \sqrt{\Psi(n_K)/\Psi(n_C)}$.

(iii) Classification. *Under the conditions of (ii), if* (21) *holds and* $a_K R_{\text{speed}} \geq 1$*, the carbon-minimizing design is faster and more dispersed; if* (21) *is reversed and* $a_K R_{\text{speed}} \leq 1$*, it is slower and more consolidated. Whatever the density, it is faster if* $a_K R_{\text{speed}} > \overline{R}$ *and slower if* $a_K R_{\text{speed}} < \underline{R}$*. Only when the two conditions point in opposite directions and* $\underline{R} \leq a_K R_{\text{speed}} \leq \overline{R}$ *does the direction of speed depend on the size of the density change;* (22) *then determines it. At equality in* (21)*, the direction of speed is that of* $a_K R_{\text{speed}}$ *relative to one.*

Proposition 7 answers, for a given product, the two questions a firm asks of its network. The density condition (20) is Proposition 5 with one change. Adding a distribution center incurs a facility charge and lost pooling, in the proportions $1 - \omega$ and ω of its marginal cost. The carbon-minimizing firm holds the lost pooling at its own speed $\hat{v}_C$, and safety stock falls with the square root of speed, so per dollar of lost pooling its carbon is $r_h \sqrt{v_K/\hat{v}_C}$. The carbon-minimizing network is more dispersed when a dollar of last-mile delivery emits more than this blend of facility and lost-pooling carbon, whether or not speed is at a bound. When the speeds are interior, the carbon-minimizing speed at the cost-minimizing density, $\hat{v}_C$, differs from v_K by the factor $(a_K R_{\text{speed}})^{2/5}$, and the lost-pooling carbon becomes the weighted geometric mean in (21): along the carbon-minimizing speed response, an additional center raises holding carbon by four-fifths of the carbon of lost pooling and inbound freight carbon by one-fifth. One consequence has no counterpart in the marginal analysis: the carbon intensity of inbound freight bears on how many distribution centers to operate, through the speed at which the carbon-minimizing firm holds the lost pooling. The effect is modest. It enters with the pooling weight and the exponent one-fifth, and it vanishes where the carbon-minimizing speed stays at a bound. Financing stock in transit raises the cost-minimizing speed, and with it the carbon of lost pooling, so it raises the threshold for dispersion. The exponents are these shares of lost-pooling carbon, and without financing also the cost shares: with freight cost and carbon proportional to v^p rather than v^2, holding's share, and with it the exponent on r_h, becomes $2p/(2p+1)$, and inbound freight's becomes $1/(2p+1)$, as does the magnitude of the exponent on a_K.

The speed decomposition (22) separates the two reasons a carbon-minimizing chain replenishes at a different speed. The intensity effect, $\frac{2}{5}\ln(a_K R_{\text{speed}})$, is the speed reversal of Proposition 3. The density effect, $\frac{1}{5}\ln[\Psi(n_C)/\Psi(n_K)]$, is complementarity: a more dispersed network holds more safety stock and so values speed more. The bands (19) bound the density effect, because no network pools more than full consolidation or less than full dispersion. Whether speed then changes also depends on whether either objective finds speed worth buying within the bounds.

Corollary 8 (Magnitudes without pooling)**.** *Under the quadratic instantiation, in the no-pooling limit (*$\rho \to 1$*),* $\omega = 0$*, the two decisions separate, and for interior optima*

$$\frac{n_C}{n_K} = R_{\text{density}}^{2/3}, \qquad \frac{v_C}{v_K} = \left(a_K R_{\text{speed}}\right)^{2/5}.$$

Carbon separates into a proximity component $C_{\text{prox}}(n) = \varphi n + \varepsilon_L N \mu \delta n^{-1/2}$ *and an availability* *component* $C_{\text{avail}}(v) = \eta z \sigma N \sqrt{D/v} + \varepsilon_0 N \mu D\, v^2$, *and the carbon of the cost-minimizing design relative to the carbon minimum* $C_{\min}$ *is*

$$\frac{C(n_K, v_K)}{C_{\min}} = \theta\,\frac{x + 2x^{-1/2}}{3} + (1-\theta)\,\frac{4y^{-1/2} + y^2}{5}, \qquad x = R_{\text{density}}^{-2/3}, \quad y = \big(a_K R_{\text{speed}}\big)^{-2/5}, \quad (23)$$

where $\theta = C_{\text{prox}}(n_C)/C_{\min}$ *is the proximity share of the minimum. To second order in the logarithms of the ratios, the excess carbon of the cost-minimizing design is* $\theta\,(\ln R_{\text{density}})^2/9 + (1-\theta)\,\frac{2}{25}\big[\ln(a_K R_{\text{speed}})\big]^2$.

Corollary 8 gives the size of the redesign. Absent pooling and with both optima interior, a product whose last mile is four times as carbon-intensive per dollar as its facilities runs $4^{2/3} \approx 2.5$ times as many distribution centers when minimizing carbon. To second order, the excess carbon of the cost-minimizing design grows with the square of the logarithm of each ratio: it is small near parity and rises quickly away from it. The cost-minimizing network emits about 5% more proximity carbon than the minimum when $R_{\text{density}} = 2$ and about 19% more when $R_{\text{density}} = 4$. Carbon-aware redesign therefore matters most for products whose ratios lie far from one. With pooling the two decisions interact and these expressions are no longer exact; Section 5 illustrates the magnitudes in that case.

The carbon-price path. A firm facing a rising carbon price needs to know how the design moves along the way, not only where it ends. For an activity with carbon intensity per dollar r_x, we use $s_x(\tau) = \tau r_x/(1+\tau r_x)$ to describe the *carbon share* of its blended rate — for holding, $s_h = \tau\eta/(h+\tau\eta)$. The share rises from zero to one as the price rises.

Proposition 9 (Carbon-price path). *Under the quadratic instantiation:* (i) *at a fixed density and an interior speed response,* $\mathrm{d}\ln v^*/\mathrm{d}\ln\tau = (a_\tau s_h - s_t)/(3 - a_\tau/2)$, *where* $a_\tau = H_\tau \mathrm{SS}/(H_\tau \mathrm{SS} + 2h_p P)$ *at the optimum; its sign is that of* $a(n, v^*)\, r_h - r_t$, *so the speed moves monotonically in* τ, *and without financing the elasticity is* $\frac{2}{5}(s_h - s_t)$; (ii) *in the no-pooling limit and for an interior density,* $\mathrm{d}\ln n^*/\mathrm{d}\ln\tau = \frac{2}{3}\,(s_L - s_F)$; (iii) *without financing, with pooling and interior speeds, if* r_L *exceeds each of* r_F, r_h, *and* r_t, *the optimal density* $n^*(\tau)$ *is nondecreasing along the whole price path, and if* r_L *falls short of each, it is nonincreasing.*

The carbon share of a blended rate measures how much of that activity's cost, carbon charge included, is carbon. By part (i), at a given network and an interior speed, a rising carbon price speeds replenishment exactly when the inventory saving from speed is more carbon-intensive per dollar than inbound freight; without financing, exactly when carbon is a larger share of the cost of holding than of inbound freight. The response is gradual: at a low price both shares are near zero and speed barely moves, while at a high price both approach one and speed settles at its carbon-minimizing level. Part (iii) gives a condition a firm can check once. Without financing, when its last mile is the most carbon-intensive of the four activities per dollar and speeds are interior, no increase in the carbon price calls for consolidation; when the last mile is the least carbon-intensive, no increase calls for dispersion. Outside these two cases, part (iii) does not fix the direction along the whole path.

Demand uncertainty: a chance-constrained cap. The cost and carbon functions are expected per-period flows, so the cap in (9) bounds *expected* carbon. Because demand is random, realized carbon fluctuates about $C(n,v)$: the facility charge and the base-stock safety-stock target are planned and treated as deterministic, but inbound and last-mile freight scale with realized throughput $Q = \sum_i X_i$, the sum of the realized market demands, which has mean $N\mu$ and variance $\sigma_Q^2 = \sigma^2 N\,[\,1 + (N-1)\rho\,]$. Realized carbon is then

$$\tilde{C}(n,v) \;=\; \varphi n + \eta\,\mathrm{SS}(n,v) + \big[\varepsilon(v)D + \varepsilon_L d(n)\big]\,Q, \tag{24}$$

and a firm that requires realized carbon to stay within B with confidence $1-\beta_c$ — where β_c is the carbon-side tail probability, denoted with the subscript c to distinguish it from the inventory fill-rate β of the lot-sizing literature — imposes the chance constraint $\Pr\{\tilde{C}(n,v) \le B\} \ge 1-\beta_c$, the carbon-side counterpart of the cycle service level α on the inventory side.

Proposition 10 (Deterministic equivalent of the chance constraint)**.** *This chance constraint is equivalent to*

$$\varphi n + \eta\,\mathrm{SS}(n,v) + m\big[\varepsilon(v)N\mu D + \varepsilon_L N\mu\, d(n)\big] \;\le\; B, \tag{25}$$

the carbon function (7) *with the transport component scaled by* $m = q_{1-\beta_c}/(N\mu)$*, where* $q_{1-\beta_c} = \inf\{q : \Pr(Q \le q) \ge 1-\beta_c\}$ *is the lower* $(1-\beta_c)$ *quantile of realized throughput* Q*, and the facility and holding terms unchanged. Under the normal approximation* $Q \sim \mathcal{N}(N\mu, \sigma_Q^2)$ *this is* $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$*, with* $z_{\beta_c} = \Phi^{-1}(1-\beta_c)$ *and* $\mathrm{CV}_Q = \sigma_Q/(N\mu)$*; then* $m \ge 1$ *whenever* $\beta_c \le \frac{1}{2}$*.*

The chance constraint therefore leaves the structure intact: (25) is the four-component carbon function with the inbound and last-mile carbon rates scaled, $\varepsilon_0 \mapsto m\varepsilon_0$ and $\varepsilon_L \mapsto m\varepsilon_L$, while cost and the facility and holding terms are unchanged. Because Q is realized market throughput, exogenous to the firm's choice of n and v, the factor m is a decision-invariant scalar on the transport carbon rates; the priced objective is again (10) under this substitution, so Propositions 2–9 carry over with ε_0 and ε_L replaced by $m\varepsilon_0$ and $m\varepsilon_L$, and only the thresholds shift.

Corollary 11 (Uncertainty tightens the speed threshold and relaxes the density threshold)**.** *Under the chance-constrained cap the speed threshold* (13) *becomes* $a(n, v_{\mathrm{K}})\,\eta/h > m\,\varepsilon'(v_{\mathrm{K}})/t'(v_{\mathrm{K}})$ *(*$m\,\varepsilon_0/t_1$ *under the quadratic), and the pooling-adjusted density threshold* (16) *becomes*

$$m\,\frac{\varepsilon_L}{t_L} \;>\; \frac{\varphi + \eta\,G\big(n_K(v), v\big)}{F + h\,G\big(n_K(v), v\big)}, \tag{26}$$

which reduces to $m\,\varepsilon_L/t_L > \varphi/F$ *in the no-pooling limit (*$\rho \to 1$*); in the joint design the speed ratio becomes* $a_K R_{\mathrm{speed}}/m$ *and the density condition* (21) *becomes*

$$m\,r_L > (1-\omega)\,r_F + \omega\, m^{1/5} r_h^{4/5} r_t^{1/5} a_K^{-1/5}.$$

Under the normal approximation, $m \geq 1$ for $\beta_c \leq \frac{1}{2}$, with equality at $\beta_c = \frac{1}{2}$, so the chance constraint tightens the speed-reversal threshold and relaxes the dispersion threshold relative to the expected-value cap, and the effect grows as m rises with the demand coefficient of variation σ/μ, the correlation ρ, and the confidence $1 - \beta_c$. Under the quadratic instantiation, in the joint design the carbon-minimizing speed is nonincreasing in m, and strictly decreasing where n_C and v_C are interior.

Demand uncertainty therefore affects carbon through two channels. It enters the inventory-holding component through the safety stock $z\sigma\sqrt{L}$, the channel behind the speed reversal, and, under a chance-constrained cap, it enters the transport component through the margin $z_{\beta_c}\,\mathrm{CV}_Q$ that the firm holds against realized emissions just as it holds safety stock against demand. More variable or more correlated demand raises this margin and, like a higher service requirement, shifts the carbon-minimizing configuration; the expected-value cap is the limiting case $m = 1$. Carrying the realized variability of on-hand inventory as well would place an analogous margin on the inventory-holding component; we keep the safety stock at its planned level, consistent with the base-stock policy. The deterministic equivalent itself is distribution-free in the throughput quantile $q_{1-\beta_c}$; the normal approximation enters only through the closed form $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$, which is most accurate for moderate N and modest CV_Q and which a firm with strongly skewed demand can replace by the empirical throughput quantile without altering the structure.

The cap applies over the same accounting period as the flow model — here a week. For a cap on cumulative carbon over T periods with demands independent across periods, aggregate throughput has coefficient of variation $\mathrm{CV}_Q/\sqrt{T}$, so the multiplier tightens toward one as the horizon lengthens: in the calibrated network of Section 5, the 95% multiplier falls from $m = 1.37$ for a weekly cap to $m = 1.05$ for an annual cap ($T = 52$). The chance-constrained results are therefore most relevant to high-frequency operational carbon budgets, shipment-cycle accounting, and corporate carbon controls, and less binding for annual compliance accounting.

Load-factor sensitivity. Our model treats the per-unit last-mile rates t_L and ε_L as independent of shipment size. Last-mile cost and carbon, however, accrue largely per *vehicle*-mile — driver time and fuel — so the per-unit rates scale with vehicle-miles per unit, which rise as dispersion thins each territory's dispatch volume and average vehicle fill falls; load-sensitive rates therefore introduce a consolidation force outside the constant-rate model. Because a fill multiplier depends on n, it enters the marginal density condition through both the level and the derivative of the last-mile term. When a common fill multiplier is applied to the cost and the carbon rate alike, both optima shift toward consolidation, while their relative direction remains consistent with the carbon-intensity comparison. Per vehicle-mile, cost is dominated by driver time while fuel use falls somewhat in an emptier vehicle — the basis of tonne-kilometer emission accounting (Smart Freight Centre 2022) — so when fill deteriorates, the per-unit cost rate typically inflates at least as strongly as the per-unit carbon rate. A multiplier falling more heavily on carbon is therefore a stress case for the dispersion result; under it, the density comparison can shift toward consolidation. A full treatment would make shipment size, routing, and consolidation additional decisions; we leave it as an extension.

5 A Calibrated Application

This section applies the results to distribution activities measured from public US data. Section 5.1 constructs benchmark carbon intensities per dollar from public sources and stated assumptions, Section 5.2 traces how a carbon price changes the speed and density of a reference network, and Section 5.3 examines whether that response survives changes in the assumptions. The directions follow from the propositions once the ratios are known, subject to the bounds on speed; the magnitudes describe the network we calibrate.

5.1 Benchmark carbon intensities

Each ratio is the carbon an activity emits per dollar it costs. We compute it from the bottom up, as the activity's energy use times the emission factor of that energy, divided by the activity's cost, and compare the result with the industry factors of the US environmentally extended input–output model (U.S. Environmental Protection Agency 2024a). A dollar spent on energy carries about 2.8 kg CO_2e as diesel, 2.9–4.6 kg as grid electricity, and 5.0–5.8 kg as marine fuel, while the labor and capital that make up the rest of logistics cost carry little carbon of their own. An activity's carbon intensity per dollar is therefore close to its energy share of cost times a few kilograms.

Public data report the energy of a whole warehouse, whereas the model separates a facility's baseline energy, φ, from the increment that varies with the inventory held, η. We split building energy by a share β that varies with inventory, and the storage charge of a third-party warehouse by a share γ that is variable for an owned facility; we take $\beta = \gamma = \frac{1}{2}$. The two shares divide one building's energy and cost, so each is counted once. Storage emissions per pallet depend on the storage environment and not on the value of the goods. We treat site energy as electricity, which overstates carbon for buildings partly heated by gas. Table 2 reports the ratios.[2]

Without spoilage, holding carries little carbon per dollar, because its cost is mostly the capital tied up in stock and the space and labor to store it, and the ratio falls with value density. Speed

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[2] Sourced inputs. Emission factors: diesel 10.21 kg CO_2 per gallon and the US grid 0.371 kg CO_2e per kWh (U.S. Environmental Protection Agency 2022); marine fuel 3.114 t CO_2 per tonne (European Commission 2016). Prices: diesel \$3.66 per gallon (2025 average) (U.S. Energy Information Administration 2026), electricity 8.1–12.8 cents per kWh (U.S. Energy Information Administration 2025), and marine fuel at average prices of \$535 (2025) and \$626 (2024) per tonne (Ship & Bunker 2026). Warehouses: site energy of 22.7 (ambient) and 84.1 (refrigerated) kBtu per square foot (U.S. Environmental Protection Agency 2024b); about 30 (ambient) and 109 (refrigerated) kg CO_2e per pallet position-year (Lineage, Inc. 2024); industrial rent of \$10.12 per square foot (Cushman & Wakefield 2025). Line haul: truckload at \$2.26 per mile, of which fuel is \$0.48 (American Transportation Research Institute 2025). Last mile: the courier and local-trucking industry factors (U.S. Environmental Protection Agency 2024a) and electric-van energy use of 0.37–0.39 kWh per mile (Mulholland 2022). Food: input–output intensities of processed beef, fluid milk, and poultry (U.S. Environmental Protection Agency 2024a). Assumptions. Third-party storage at \$240 (ambient) and \$480 (refrigerated) per pallet-year; rent at about 40% of a distribution center's fixed operating cost, with cold-storage rent twice ambient; capital at 10% of pallet value, for stock held and stock in transit; intermodal 10–30% cheaper than truckload with about 60% less carbon per load; a team premium of about \$0.35 per mile, carrying about 0.03 kg CO_2e per dollar, at twice the daily distance of a solo driver; air at four to five times the cost of road; pallet values; and the allocation shares β and γ. Ranges combine the low and high values of every input.

Table 2: Benchmark carbon intensity per dollar of distribution activities, kg CO_2e per US dollar, with ranges over the stated inputs and allocation shares.

Ratio	Activity	Central	Range
Holding, r_h	Ambient packaged goods	0.056	0.007–0.32
	High-value goods	0.003	0.000–0.015
	Refrigerated dairy	0.149	0.021–0.62
	Refrigerated dairy, spoilage 0.5% per week	0.457	0.20–0.73
Line-haul speed, r_t	Faster vessel speed	5.37	4.97–5.82
	Truck instead of intermodal rail	1.78	1.18–3.55
	Air instead of truck	1.10	0.60–1.90
	Team instead of solo drivers	0.03	0.00–0.10
Last mile, r_L	Diesel local truck	0.50	0.30–0.60
	Diesel van	0.30	0.15–0.45
	Electric truck	0.16	0.12–0.20
	Electric van	0.075	0.05–0.10
Facilities, r_F	Ambient distribution center	0.049	0.015–0.11
	Refrigerated distribution center	0.091	0.027–0.20

bought with fuel — a faster vessel, truck instead of intermodal rail, air instead of truck — is the most carbon-intensive, because the added cost is mostly fuel. A team of two drivers runs the same tractor about twice as far per day, so the speed it buys adds mostly wages and little carbon. Spoilage changes the holding ratio of perishables: food that spoils while held carries the carbon of its production, which we value at the input–output intensity of the food. Even beef, the highest of the food categories we examine at 1.12 kg per dollar of processed product, stays below the ratio of buying speed with truck instead of intermodal rail or with vessel speed; air freight, whose ratio spans 0.60 to 1.90, is the one fuel-bought option against which spoilage could favor speed. The spoilage rate of 0.5% of stock value per week is an assumption.

5.2 How a carbon price changes a reference network

We calibrate the model to a distributor that replenishes stores in forty metropolitan markets with ambient goods worth \$50,000 a pallet and that owns the goods while they travel. Table 3 maps each input of the model to its counterpart in this network. The firm can replenish at any speed between intermodal rail, at 250 miles a day, and truckload, at 500; we fit $t(v)$ and $\varepsilon(v)$ through these two services. The fitted carbon rate includes a constant term, which raises reported carbon but leaves every decision unchanged because throughput is fixed.

At the cost minimum the network operates four distribution centers and ships in 2.15 days, close to truckload, because financing makes speed valuable: the stock in transit, 3,065 pallets, exceeds the safety stock, 2,216, so the safety-stock share is $a_K = 0.27$. The pooling weight is $\omega = 0.08$. Holding emits 0.003 kg per dollar and the speed that truckload buys over rail 1.78, so the carbon pressure on speed points toward slower replenishment (Proposition 3). Diesel delivery emits 0.50 kg per dollar, far above the threshold of 0.046 in Proposition 7(i), so the carbon pressure on density

Table 3: Inputs of the reference network.

Input	Model	Reference network
Markets and demand	N; μ, σ; ρ	40 markets; 250 pallets a week each, coefficient of variation 0.40; correlation 0.30
Service	z	95% cycle service ($z = 1.645$)
Distances	D; δ	1,000 miles inbound; $d(n) = 700\, n^{-1/2}$ miles to the markets
Line haul	v; $t(v)$; $\varepsilon(v)$	250 (rail) to 500 (truckload) miles a day; \$0.070 to \$0.087 and 0.021 to 0.052 kg per pallet-mile
Store delivery	t_L; ε_L	diesel truck, \$2.75 a mile with 20 pallets a load; $r_L = 0.50$
Distribution center	F; φ	\$3.0 million and 148 t a year
Holding	h; η	\$5,120 and 15 kg per pallet-year
Stock in transit	h_p	\$5,000 per pallet-year, 10% of the value of the goods

points toward more distribution centers.

Figure 2 traces the cost-minimizing design as the carbon price rises, for three sets of actions: changing replenishment alone, changing the number of distribution centers alone, and changing both. The price moves speed first. Up to about \$570 per tonne the network keeps four centers while the line haul slows toward rail. At \$100 per tonne emissions fall by 5.4% at a cost increase of 0.1%, and at \$500 per tonne by 17.9% at a cost increase of 1.3%; the last tonne avoided costs the carbon price, and the average tonne costs \$48 and \$205, respectively. Speed adjusts continuously, whereas the network changes by whole centers: with the number of centers treated as continuous, the network grows by about half a center by \$500 per tonne, but a fifth center pays only above about \$570. The network then grows to five centers at \$1,000 per tonne, six at \$2,000, and eight at \$5,000, while the line haul reaches rail at about \$3,000. Changing the network alone achieves little: at \$2,000 per tonne it lowers emissions by 5.4%, against 32.6% for replenishment alone and 37.9% for both.

In the reference network, the carbon intensity of faster service does not change the carbon-minimizing number of centers. Whether faster service emits 0.03, 0.45, or 1.78 kg per dollar, the carbon-minimizing chain ships by rail and operates 19 centers, because the holding carbon that speed would save is smaller still. The link of Proposition 7 requires speed to respond. For refrigerated dairy with spoilage, served by solo or team drivers, with more variable demand and smaller centers (coefficient of variation 0.60, correlation 0.1, 99% service, and centers costing \$1.0 million a year, for a pooling weight of 0.10), the carbon-minimizing network has 23 centers when faster service emits 0.03 kg per dollar and 21 when it emits 1.78.

5.3 Robustness

Table 4 examines whether the response to a carbon price of \$1,000 per tonne, at which both decisions move, survives changes in the assumptions. The first two rows are implementation checks, the next five are parameter and accounting sensitivities, and the last is a model extension. Each row reports the cost-minimizing design and the design at that price, with the emissions cut and the cost increase measured from the row's own cost minimum. For the caps on realized carbon, the budget is the

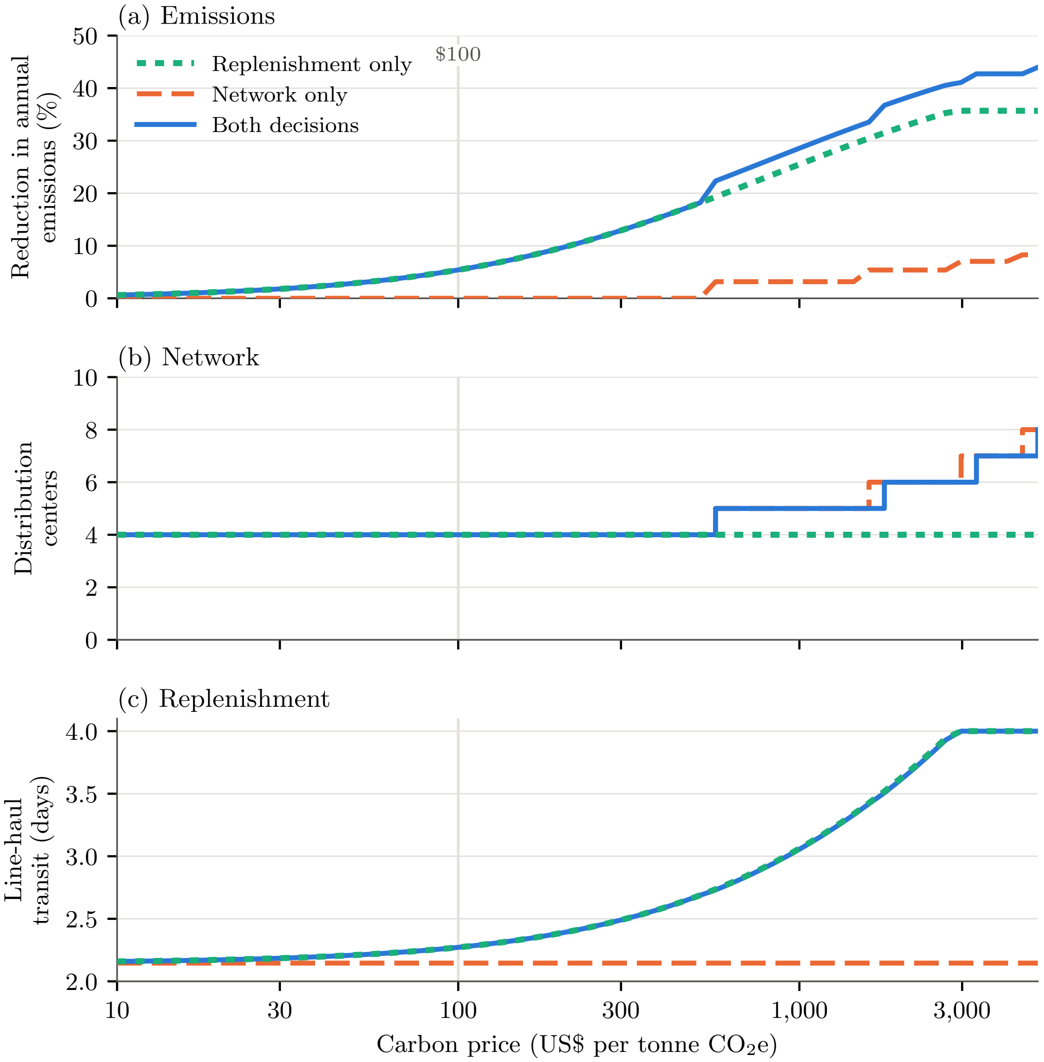

Figure 2: Cost-minimizing design of the reference network as the carbon price rises, for three sets of actions.

expected carbon of the reference design at that price, and demand is independent across weeks.

In every row the carbon price slows replenishment and adds a distribution center or more. Restricting speed to the two services removes the intermediate speeds, so the cost-minimizing chain uses truckload and the carbon price moves it to rail, with a larger cut at a larger cost: 43.2% at 9.7%. Treating the number of centers as continuous changes the design by a fraction of a center and the cost increase by 0.1 percentage point. The share β of building energy that varies with stock changes the cut by less than one percentage point. Without financing, the cost-minimizing chain already ships near rail, at 3.62 days, so the same price cuts emissions by 9.4% at a cost increase of 0.9%. A cap on realized rather than expected carbon, at the same average budget, requires a deeper cut: for an annual cap, 31.9% with five centers at a cost increase of 5.1%; for a weekly cap, 46.6% with twelve centers at a cost increase of 24.0%. In the extension, last-mile cost and carbon

Table 4: Robustness of the response to a carbon price of \$1,000 per tonne. Days are line-haul transit times.

	Cost minimum		At \$1,000 per tonne			
	Centers	Days	Centers	Days	Emissions cut	Cost increase
Reference	4	2.15	5	3.06	28.6%	3.8%
Implementation checks						
Rail or truckload only	4	2.00	5	4.00	43.2%	9.7%
Continuous number of centers	3.9	2.15	4.9	3.06	28.6%	3.7%
Parameter and accounting sensitivities						
Building energy share $\beta = 0.25$	4	2.15	5	3.06	28.2%	3.8%
Building energy share $\beta = 0.75$	4	2.15	5	3.05	29.0%	3.8%
No financing of stock in transit	4	3.62	5	4.00	9.4%	0.9%
Annual cap on realized carbon, 95%	4	2.15	5	3.29	31.9%	5.1%
Weekly cap on realized carbon, 95%	4	2.15	12	3.98	46.6%	24.0%
Model extension						
Load-sensitive delivery, $\lambda = 0.25$	2	2.16	3	3.07	27.2%	3.6%

per pallet scale with $(n/4)^\lambda$, so that vehicle fill falls as the network disperses; with $\lambda = 0.25$ the cost-minimizing network has two centers and the carbon price adds one. With $\lambda = 0.1$ the network moves from three centers to four, and with $\lambda = 0.5$ it stays with one center while the line haul slows to 3.08 days.

6 Managerial Implications

The results give a firm a diagnostic it can compute from its own rate data, and they indicate which of its distribution decisions a carbon price will move, in which direction, and by how much. Each implication below follows from a result of Section 4, and the calibrated values of Section 5 illustrate it.

A diagnostic in two questions. A firm needs four ratios and two shares. The ratios are the carbon per dollar of holding stock, of the speed it can buy on its line-haul lane, of last-mile delivery, and of operating a distribution center. Each is close to the activity's energy share of cost times the carbon of a dollar of its energy, together with, for perishables, the production carbon of stock that spoils (Section 5.1), so it can be estimated from fuel and electricity spend and spoilage records. The shares are the pooling weight ω — the share of a distribution center's marginal cost that is lost pooling — and the safety-stock share a_K of the inventory that speed saves, and the firm can obtain both from its cost-minimizing network through (18) and (8). Proposition 7 then answers two questions: whether replenishment should be faster or slower, and whether the network should have more or fewer distribution centers. Table 5 sets out the four answers; off the diagonal, the speed direction shown requires $a_K R_{\text{speed}}$ to lie outside the bands (19).

Table 5: Carbon-minimizing direction of the two decisions (Proposition 7), with examples from the central ratios of Table 2.

	More distribution centers *last mile more carbon-intensive per dollar than facilities and lost pooling*	**Fewer distribution centers** *last mile less carbon-intensive per dollar than facilities and lost pooling*
Faster replenishment *inventory saving from speed more carbon-intensive per dollar than the speed it buys*	team drivers for perishables that spoil; diesel delivery or electric trucks	team drivers for perishables that spoil; electric vans serving refrigerated centers
Slower replenishment *inventory saving from speed less carbon-intensive per dollar than the speed it buys*	speed bought with fuel, or team drivers for high-value goods; diesel delivery, electric trucks, or electric vans serving ambient centers	speed bought with fuel; electric vans serving refrigerated centers

Slow the speed that is bought with fuel. A faster vessel, truck instead of intermodal rail, and air instead of truck add cost that is mostly fuel, so their carbon per dollar is many times that of holding stock (Table 2). Proposition 3 then calls for slower replenishment, and financing the goods in transit strengthens that call: it makes the cost-minimizing chain buy speed for the capital it saves, which carries no carbon, and so lowers the share a_K. In the reference network, which finances its stock in transit, a carbon price slows the line haul at every price, and up to about \$570 per tonne it changes nothing else (Figure 2). The shifts of pharmaceutical freight from air to ocean and of truckload lanes to intermodal rail are consistent with this direction (Pharmaceutical Commerce 2023; FreightWaves 2026), as is the finding of Maloni et al. (2013) that slower vessel speed lowers carbon and total cost despite the added pipeline inventory. Where the cost-minimizing chain already ships at the slowest option, a carbon price leaves speed there.

Buy speed with labor where held stock carries carbon. Speed bought with labor, such as a team of drivers on the same tractor, adds wages rather than fuel, so its carbon per dollar is close to zero, and a carbon price favors it wherever the inventory saving from speed is more carbon-intensive per dollar than that speed (Proposition 3). That requires held stock to carry carbon, as it does for perishables that spoil, whose holding ratio is fifteen times that of team operation (Table 2). For high-value goods, whose holding cost is mostly capital, team operation is the more carbon-intensive of the two, and a carbon price slows it. Financing goods in transit narrows the case for faster replenishment, because it lowers the share a_K. Demand volatility and the service target raise the cost-minimizing and the carbon-minimizing speeds together; without financing they leave the direction unchanged, and with financing they raise a_K and can turn it toward faster replenishment, never slower (Proposition 4).

Reassess network density when delivery technology changes. The density decision compares the carbon per dollar of last-mile delivery with that of a distribution center's facility and lost pooling (Propositions 5 and 7). Diesel delivery, by truck or van, and electric trucks are carbon-intensive enough per dollar to call for more distribution centers closer to demand. The electric van lies close to the threshold: at the central ratios of Table 2 it calls for more distribution centers when they are ambient and for fewer when they are refrigerated, and its direction also depends on how much of a building's energy the facility carries. Amazon reports that placing inventory closer to customers avoided 127 million vehicle miles in 2025 (Amazon.com, Inc. 2026), a move in the direction Proposition 7 gives when the last mile is carbon-intensive. A firm that electrifies its last mile should therefore revisit how many distribution centers it operates. A firm that adds centers to shorten delivery should weigh the vehicle fill it gives up, since emptier vehicles raise both the cost and the carbon of each delivered unit (Section 5.3). The carbon intensity of the line haul bears on density only through the speed at which the carbon-minimizing firm holds the lost pooling, and the effect is modest; in the reference network it does not change the carbon-minimizing number of centers (Section 5.2).

Expect the network to move only at high carbon prices. Proposition 9 ties the response to a carbon price to the carbon share of each activity's cost, carbon charge included. At \$100 per tonne, carbon is under 1% of the cost of holding ambient goods and of operating facilities, 5% of the cost of diesel truck delivery, 15% of speed bought with truck instead of rail, and 35% of speed bought with vessel speed; it reaches half of the cost of diesel truck delivery only at about \$2,000 per tonne. In the reference network a price of \$100 per tonne lowers emissions by 5.4% at a cost increase of 0.1%, through slower line haul alone, and the number of centers changes only above about \$570 per tonne (Figure 2). Changing replenishment and the network together achieves more than either alone: at \$2,000 per tonne, 37.9% against 32.6% and 5.4%. The two decisions are complements (Proposition 6), so a program that acts on one alone, such as a slow-steaming mandate or a consolidation drive, evaluates it at a level of the other decision that the joint design would change.

Account for realized carbon over short horizons. A cap on realized rather than expected carbon places a risk margin on transport emissions (Proposition 10). For a weekly cap held with 95% confidence, the margin raises the transport carbon rates by 37% in our calibration and so reinforces slower fuel-bought speed and, with diesel delivery, more distribution centers (Corollary 11). At the budget the reference network reaches at \$1,000 per tonne, a weekly cap calls for twelve centers rather than five (Table 4). For an annual cap the margin is about 5% and changes the design less.

Implications for policy. An instrument that prices or caps transport alone, such as a freight fuel levy or a vehicle emissions standard, blends the carbon price into the transport rates and leaves holding and facility carbon unpriced. At a given density it therefore slows replenishment wherever speed is above its lower bound, and at a given speed it adds distribution centers, whatever the product. By the marginal comparisons of Propositions 3 and 5, these moves lower total carbon, at least for small prices, when they agree with the carbon-minimizing direction, and raise it when they do not. At a given density, a transport-only price therefore raises total carbon when the inventory saving from speed is more carbon-intensive per dollar than the speed it would buy; at a given speed, it does so when the last mile is less carbon-intensive per dollar than facilities and lost pooling. When both decisions adjust, its effect on total carbon depends on their joint response. By the ratios of Table 2, the first condition holds for perishables that spoil served by team drivers, where the chain buys such speed, and the second for electric vans serving refrigerated centers. An instrument that prices total logistics carbon, including warehousing, lets the firm choose the speed and density that minimize carbon across all four activities.

7 Discussion and Conclusion

A firm that distributes through regional distribution centers provides availability through safety stock or replenishment speed and proximity through facilities or last-mile travel, and a carbon price or cap can change which activity provides each. We have shown that the joint design has a unique optimum at every carbon price and that the carbon-minimizing direction of both decisions follows from four carbon-intensity-per-dollar ratios and two shares of the cost-minimizing design: the share of a distribution center's marginal cost that is lost pooling, and the share of the inventory saving from speed that is safety stock. A firm that finances its goods in transit buys speed for capital that carries no carbon, so a carbon price works toward slower line haul. The carbon intensity of line-haul freight enters the density decision through the speed that recovers lost pooling, and the effect is modest. In public US data, the classification depends on the product, on how speed is bought, and on how the last mile is driven. At the central ratios, speed bought with fuel is slowed for every product, and speed bought with labor is favored where held stock carries carbon, as for perishables that spoil; diesel delivery and electric trucks call for more distribution centers, and electric vans for fewer when the centers are refrigerated. In a reference network that finances its stock in transit, a carbon price slows the line haul at every price and changes the number of distribution centers only at high prices or under stringent caps. Several documented moves in practice are consistent with these directions; in the sources we examine, firms report them in terms of cost, delivery service, and transport miles. Our model adds the carbon of the inventory a slower chain holds, the financing of the goods it carries, and the link between speed and density.

Limitations. The model describes an average, symmetric distribution environment, so its classification is a diagnostic for a heterogeneous network rather than a design for it. Heterogeneous markets or discrete facility sites replace $d(n) = \delta n^{-1/2}$ with the realized last-mile profile; this can move the density threshold but leaves the comparison between facility and last-mile carbon per dollar intact. The representative inbound distance isolates the proximity effect of adding distribution centers. If more centers also mean smaller inbound shipments, lower vehicle fill, or lost facility scale, those forces favor consolidation and can move a case near the threshold, as last-mile fill does (Section 5.3). We charge the financing of goods in transit to the firm that owns them; a firm that buys on delivered terms may still bear that cost through the supplier's price. We treat speed as a continuous index between real options on a lane. A firm that splits volume between modes faces a mixture of lead times, and our safety-stock expression does not represent such a mixture. The model ties last-mile distance to the number of stocking distribution centers, whereas networks that deliver through cross-docks or hubs separate stocking density from delivery density. Service enters as a cycle-service requirement; if it is chosen, safety stock becomes a further design margin. Finally, the carbon intensities are calibrated to public industry data rather than to any firm's records, so the classification should be computed with a firm's own rates, and the magnitudes in Section 5 describe the network we calibrate.

Future research. Multi-product and multi-echelon settings would show how a shared network serves products in different regimes and where safety stock should be held. Separating stocking from delivery density, and modeling shipment consolidation and routing explicitly, would quantify the centralizing forces represented here by fill sensitivity. Accounting under ISO 14083 makes firm-level logistics-emissions data increasingly available (International Organization for Standardization 2023); such data would locate real products relative to the speed and density thresholds. Endogenous service choice and explicit mode mixes would make the service target and a lane's mode split design decisions rather than inputs.

Carbon accounting asks a distribution network two questions, how fast to replenish and how many distribution centers to operate, and the answer depends on which activity provides availability and proximity at the lower carbon intensity per dollar. In the US data we calibrate, at the central ratios, that comparison favors slower fuel-bought freight for every product and, where the last mile runs on diesel vehicles, more distribution centers closer to demand; in the reference network a carbon price moves the line haul well before it moves the network.

Data availability

The data and code supporting the numerical illustrations are available from the author on reasonable request.

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A Proofs

Proof of Lemma 1. The blended rates in (10) follow by collecting coefficients in $K+\tau C$; the financing term has no carbon counterpart. The speed comparison sets holding, weighted by the share a, against inbound transport at the margin (Proposition 3); the density comparison sets last-mile transport against a pooling-adjusted blend of facilities and holding (Proposition 5). The four ratios listed in the statement are therefore the basic carbon-intensity comparisons used below. ☐

Proof of Proposition 2. We give the argument for J; for C replace each blended rate by its carbon rate and set $h_p = 0$. Let $x = \ln n$ and $y = \ln v$, which range over a box, and let $k = z\sigma\sqrt{ND}$. Then

$$J = F_\tau e^x + H_\tau k\, e^{g(x,y)} + h_p N\mu D\, e^{-y} + T_\tau(e^y)\, N\mu D + \ell_\tau N\mu\delta\, e^{-x/2},$$

with $g(x,y) = \frac{1}{2}\ln\big(N\rho + (1-\rho)e^x\big) - \frac{1}{2}y$. (i) The facility and last-mile terms are strictly convex in x. The function g is convex, because $\ln(c_1 + c_2 e^x)$ is convex for $c_1 \geq 0$ and $c_2 > 0$, so the holding term is log-convex and therefore convex. The transport term is convex in y, because T_τ is convex and nondecreasing and e^y is convex. For strict convexity, let $S = H_\tau k\, e^g$; its Hessian is $S\big(\nabla^2 g + \nabla g\, \nabla g^{\mathsf{T}}\big)$, with $\nabla g = (g_x, -\frac{1}{2})$ and $\nabla^2 g = \operatorname{diag}(g_{xx}, 0)$, $g_{xx} \geq 0$. The Hessian of the facility, holding, and last-mile terms therefore has (x,x) entry $F_\tau e^x + \frac{1}{4}\ell_\tau N\mu\delta e^{-x/2} + S(g_{xx} + g_x^2) > 0$, (y,y) entry $S/4$, and off-diagonal entry $-Sg_x/2$, so its determinant is $\big(F_\tau e^x + \frac{1}{4}\ell_\tau N\mu\delta e^{-x/2}\big)S/4 + S^2 g_{xx}/4 > 0$. These terms are strictly convex, and adding the convex transport and financing terms preserves strict convexity; the financing term $h_p N\mu D\, e^{-y}$ is convex in y and does not depend on x. (ii) A strictly convex continuous function has a unique minimizer on a compact convex set. Since J is continuous in (x, y, τ), the minimizer is continuous in τ by the maximum theorem. (iii) For fixed x, J is strictly convex in y, so the speed response is unique and, by the maximum theorem, continuous. The transport, financing, facility, and last-mile terms are additively separable in x and y, and the cross-partial of the holding term is $-Sg_x/2 \leq 0$, so J has decreasing differences in (x, y) and the minimizing y is nondecreasing in x. At a fixed speed, J is strictly convex in x; the reduced objective is the partial minimum of a strictly convex function over a convex set and is therefore strictly convex in x. A strictly convex function of $\ln n$ is strictly quasiconvex in n. By Danskin's theorem, which applies because the speed response is unique, J^{r} is differentiable with $J^{\mathrm{r}}_n(n) = J_n\big(n, v^*(n)\big)$, and by strict quasiconvexity J^{r}_n changes sign at most once, from negative to positive. ☐

Proof of Proposition 3. $N\mu D\, t(v)$, $N\mu D\, \varepsilon(v)$, and $h_p N\mu D\, v^{-1}$ are convex and $\kappa(n)v^{-1/2}$ has $\partial_v^2 = \frac{3}{4}\kappa(n)v^{-5/2} > 0$, so both objectives are strictly convex in v. We use $M_s = \frac{1}{2}h\,\kappa(n)\, v_{\mathrm{K}}^{-3/2} = h\mathrm{SS}/(2v_{\mathrm{K}})$ and $M_p = h_p N\mu D\, v_{\mathrm{K}}^{-2} = h_p P/v_{\mathrm{K}}$ to denote the marginal savings of safety stock and of stock in transit, so that $a(n, v_{\mathrm{K}}) = M_s/(M_s + M_p)$. The cost optimum satisfies $\hat{K}'(v_{\mathrm{K}}) = 0$, i.e. $N\mu D\, t'(v_{\mathrm{K}}) = M_s + M_p$, so that $\frac{1}{2}\kappa(n)v_{\mathrm{K}}^{-3/2} = a(n, v_{\mathrm{K}})\, N\mu D\, t'(v_{\mathrm{K}})/h$. Evaluating the carbon derivative there,

$$\hat{C}'(v_{\mathrm{K}}) = -\eta\,\tfrac{1}{2}\kappa(n)v_{\mathrm{K}}^{-3/2} + N\mu D\,\varepsilon'(v_{\mathrm{K}}) = N\mu D\big(\varepsilon'(v_{\mathrm{K}}) - a(n, v_{\mathrm{K}})\,(\eta/h)\,t'(v_{\mathrm{K}})\big),$$

which is negative — so $v_C > v_{\mathrm{K}}$ by strict convexity — if and only if (13). With $t' = 2t_1 v$ and $\varepsilon' = 2\varepsilon_0 v$ the marginal ratio is ε_0/t_1, and with $h_p = 0$ the priced first-order condition (12) gives $v^{5/2} = (h + \tau\eta)z\sigma\sqrt{ND}\,\sqrt{\Psi}/[4(t_1 + \tau\varepsilon_0)N\mu D]$, which is (14). The projection and the boundary qualification are immediate. ☐

Proof of Proposition 4. Let $s = z\sigma$, so that $\mathrm{SS} \propto s\,v^{-1/2}$. Condition (12) takes the form $M_s + M_p = 2(t_1 + \tau\varepsilon_0)N\mu D\,v$ with $M_s = H_\tau \mathrm{SS}/(2v) \propto s\,v^{-3/2}$ and $M_p = h_p N\mu D\,v^{-2}$, and let $a_\tau = M_s/(M_s + M_p)$. Implicit differentiation gives $\mathrm{d}\ln v^*/\mathrm{d}\ln s = M_s/(\frac{5}{2}M_s + 3M_p) = a_\tau/(3 - a_\tau/2) \in (0, \frac{2}{5}]$, so speed rises with service and demand volatility; without financing $v^* \propto s^{2/5}$, as in (14). Along this speed response, $J_n^{\mathrm{r}}(n; s) = J_n\big(n, v^*(n; s); s\big)$ because $J_v = 0$. The holding marginal term in (11) is proportional to $s(v^*)^{-1/2}(1-\rho)/\sqrt{\Psi(n)}$, whose elasticity in s is $1 - a_\tau/(6 - a_\tau) \geq \frac{4}{5}$, hence it rises in s even after the induced speed increase. Thus $J_{ns}^{\mathrm{r}} > 0$. By Proposition 2(iii), $J_n^{\mathrm{r}}(\cdot\,; s)$ changes sign at most once, from negative to positive, and it is increasing in s; its sign change therefore occurs at a lower n when s rises, so the optimal density is nonincreasing in s, and strictly decreasing where it is interior. Finally, at $\tau = 0$, $M_s/M_p \propto s\,v_{\mathrm{K}}^{1/2}$ rises in s, directly and through v_{K}, so $a(n, v_{\mathrm{K}})$ rises in s; under the quadratic instantiation the right side of (13) is the constant ε_0/t_1, so the comparison can change only toward faster replenishment. ☐

Proof of Proposition 5. By Proposition 2(iii), $J(\cdot, v)$ and $C(\cdot, v)$ are strictly quasiconvex on $[1, N]$, which gives uniqueness and the interior/boundary characterization. For the threshold, take $\tau = 0$ for cost and $\tau \to \infty$ for carbon. The cost density satisfies $\partial_n K = F + h\,G(n_K, v) - t_L N\mu\delta/(2n_K^{3/2}) = 0$, so the last-mile marginal $t_L N\mu\delta/(2n_K^{3/2}) = F + h\,G(n_K, v)$. Evaluating the carbon derivative there,

$$\partial_n C\big|_{n_K} = \varphi + \eta\,G(n_K, v) - \frac{\varepsilon_L N\mu\delta}{2n_K^{3/2}} = \varphi + \eta\,G(n_K, v) - \frac{\varepsilon_L}{t_L}\big[F + h\,G(n_K, v)\big],$$

which is negative — so $n_C > n_K$ by strict quasiconvexity of $C(\cdot, v)$ — if and only if (16). With $R(G) = (\varphi + \eta G)/(F + hG)$, we have $R(0) = \varphi/F$, $R(G) \to \eta/h$ as $G \to \infty$, and $R'(G) = (\eta F - h\varphi)/(F + hG)^2$, so R is monotone with $\operatorname{sign} R' = \operatorname{sign}(\eta/h - \varphi/F)$ and lies between φ/F and η/h. In the no-pooling limit $\rho \to 1$, $G \to 0$ and $\Psi \to N$, so (11) gives $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ and (16) becomes (17). ☐

Proof of Proposition 6. Since $G(n, v) \propto v^{-1/2}$, the cross-partial is $J_{nv} = H_\tau G_v$, which equals $-H_\tau G(n, v)/(2v) < 0$ for $\rho < 1$. For the speed response, where v^* is interior (12) holds; its right side strictly increases in n, through SS, and strictly decreases in v, while its left side does not depend on n and is nondecreasing in v, so v^* strictly increases in n, and projection onto the speed bounds preserves monotonicity. For the density response, $J_n(\cdot, v)$ changes sign at most once, from negative to positive (Proposition 2(iii)), and $J_{nv} < 0$; its sign change therefore occurs at a higher n when v rises. ☐

Proof of Proposition 7. (i) By Proposition 2 applied to the carbon objective, $n_C > n_K$ if and only if the reduced carbon objective slopes downward at n_K, $C^{\mathrm{r}}_n(n_K) < 0$, and $n_C = n_K$ if it is zero. By the envelope theorem (Danskin's theorem, since $\hat v_C$ is unique whether or not it is at a bound), $C^{\mathrm{r}}_n(n_K) = \varphi + \eta\, G(n_K, \hat v_C) - \frac{1}{2}\varepsilon_L N\mu\delta\, n_K^{-3/2}$. The financing term does not depend on n, so the cost condition (11) at $\tau = 0$, which holds because n_K is interior, gives $\frac{1}{2}t_L N\mu\delta\, n_K^{-3/2} = F + hG(n_K, v_K)$, and the last term equals $r_L\bigl[F + hG(n_K, v_K)\bigr]$. Since $G \propto v^{-1/2}$, $\eta\, G(n_K, \hat v_C) = h\, G(n_K, v_K)\, r_h\sqrt{v_K/\hat v_C}$. Dividing $C^{\mathrm{r}}_n(n_K)$ by $F + hG(n_K, v_K) > 0$ and substituting $\varphi = F\, r_F$ gives (20). Under the quadratic instantiation with v_K and $\hat v_C$ interior, the carbon condition (12) at n_K gives $\eta\mathrm{SS}(n_K, \hat v_C) = 4\varepsilon_0 N\mu D\, \hat v_C^2$, and the cost condition gives $h\mathrm{SS}(n_K, v_K) = 4a_K t_1 N\mu D\, v_K^2$; with SS $\propto v^{-1/2}$ these imply $(\hat v_C/v_K)^{5/2} = a_K R_{\mathrm{speed}}$, so $r_h\sqrt{v_K/\hat v_C} = r_h^{4/5} r_t^{1/5} a_K^{-1/5}$, which is (21). The second expression for ω in (18) follows from $F + hG(n_K, v_K) = \frac{1}{2}t_L N\mu\delta\, n_K^{-3/2}$ and $F = \frac{1}{2}t_L N\mu\delta\, (n_K^0)^{-3/2}$. (ii) The carbon objective has no financing term, so by (12) its speed response satisfies $v_C(n)^{5/2} \propto \sqrt{\Psi(n)}$, and $\ln[v_C/\hat v_C] = \frac{1}{5}\ln[\Psi(n_C)/\Psi(n_K)]$. Adding $\ln[\hat v_C/v_K] = \frac{2}{5}\ln(a_K R_{\mathrm{speed}})$ from part (i) gives (22). (iii) If (21) holds, then $n_C > n_K$ and $\Psi(n_C) > \Psi(n_K)$, so in (22) the first term is nonnegative and the second positive; the reversed case is symmetric. Since Ψ is increasing with $\Psi(1) \le \Psi(n_C) \le \Psi(N) = N$, the threshold $\sqrt{\Psi(n_K)/\Psi(n_C)}$ in (ii) lies in $[\underline{R}, \overline{R}]$. ☐

Proof of Corollary 8. With $\rho = 1$, $\Psi \equiv N$ and $G \equiv 0$, so (11) involves only F_τ and ℓ_τ and gives $n^*(\tau) = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$ (Proposition 5); the ratio of its carbon and cost values is $(r_L/r_F)^{2/3}$; the financing term does not depend on n. The speed response no longer depends on n, and by the argument of Proposition 7(i) the ratio of its carbon and cost values is $(a_K R_{\mathrm{speed}})^{2/5}$. For (23), a function $c_1 n + c_2 n^{-1/2}$ attains its minimum $3c_1 n^\circ$ at $n^\circ = (c_2/2c_1)^{2/3}$ and equals $c_1 n^\circ(x + 2x^{-1/2})$ at $n = xn^\circ$; with $n_K = xn_C$, $C_{\mathrm{prox}}(n_K)/C_{\mathrm{prox}}(n_C) = (x + 2x^{-1/2})/3$. Likewise $c_1 v^{-1/2} + c_2 v^2$ attains its minimum $\frac{5}{4}c_1(v^\circ)^{-1/2}$ and equals $c_1(v^\circ)^{-1/2}(y^{-1/2} + y^2/4)$ at $v = yv^\circ$, so with $v_K = yv_C$, $C_{\mathrm{avail}}(v_K)/C_{\mathrm{avail}}(v_C) = (4y^{-1/2} + y^2)/5$. Weighting by θ and $1 - \theta$ gives (23); expanding $x = e^{u_1}$ and $y = e^{u_2}$ to second order gives the last statement. ☐

Proof of Proposition 9. (i) Condition (12) is $\Upsilon(v, \tau) = 2b_\tau N\mu D\, v - M_s - M_p = 0$, with $b_\tau = t_1 + \tau\varepsilon_0$, $M_s = H_\tau \mathrm{SS}/(2v)$, and $M_p = h_p P/v$. Then $v\,\Upsilon_v = \frac{5}{2}M_s + 3M_p$ and, since $\tau\eta/H_\tau = s_h$ and $\tau\varepsilon_0/b_\tau = s_t$, $\tau\,\Upsilon_\tau = s_t(M_s + M_p) - s_h M_s$. Hence $\mathrm{d}\ln v^*/\mathrm{d}\ln\tau = -\tau\Upsilon_\tau/(v\Upsilon_v) = (a_\tau s_h - s_t)/(3 - a_\tau/2)$ with $a_\tau = M_s/(M_s + M_p)$. Substituting the definitions, $a_\tau s_h - s_t = \tau\,(h\mathrm{SS} + 2h_p P)\, t_1\, [a(n, v^*)\, r_h - r_t]/[(H_\tau \mathrm{SS} + 2h_p P)\, b_\tau]$, which has the sign of $a(n, v^*) r_h - r_t$. Because $a(n, v)$ rises with v, a positive sign raises v^* and keeps the sign positive, and a negative sign lowers v^* and keeps it negative, so the sign does not change along the path and v^* is monotone in τ. With $h_p = 0$, $a_\tau = 1$ and the elasticity is $\frac{2}{5}(s_h - s_t)$. (ii) The same computation applies to $n^* = (\ell_\tau N\mu\delta/2F_\tau)^{2/3}$. (iii) With $h_p = 0$, by (14), along the speed response $H_\tau G\bigl(n, v^*(n)\bigr) = c\,(1 - \rho)\, H_\tau^{4/5} b_\tau^{1/5}\Psi(n)^{-3/5}$, with $b_\tau = t_1 + \tau\varepsilon_0$ and $c > 0$

depending on neither n nor τ. By the envelope theorem, $J^{\mathrm{r}}_n = F_\tau + H_\tau G\big(n, v^*(n)\big) - \frac{1}{2}\ell_\tau N\mu\delta n^{-3/2}$; multiplying by $n^{3/2}/\ell_\tau > 0$ preserves its sign,

$$\frac{n^{3/2}J^{\mathrm{r}}_n}{\ell_\tau} = \frac{F_\tau}{\ell_\tau}\, n^{3/2} + c\,(1-\rho)\,\frac{H_\tau^{4/5} b_\tau^{1/5}}{\ell_\tau}\,\frac{n^{3/2}}{\Psi(n)^{3/5}} - \tfrac{1}{2}N\mu\delta.$$

This expression changes sign at most once, from negative to positive (Proposition 2(iii)). If both coefficients are nonincreasing in τ, the expression is pointwise nonincreasing in τ, so its sign change moves weakly to higher n and $n^*(\tau)$ is nondecreasing. The logarithmic derivatives of the coefficients are $1/(r_F^{-1} + \tau) - 1/(r_L^{-1} + \tau)$ and $\frac{4}{5}/(r_h^{-1} + \tau) + \frac{1}{5}/(r_t^{-1} + \tau) - 1/(r_L^{-1} + \tau)$. Since $1/(r^{-1} + \tau)$ is increasing in r, both are nonpositive at every τ when r_L exceeds r_F, r_h, and r_t, and nonnegative when it falls short of each. □

Proof of Proposition 10. In (24) the throughput coefficient $\gamma(n, v) = \varepsilon(v)D + \varepsilon_L d(n) > 0$, so $\{\tilde{C} \le B\} = \{Q \le (B - \varphi n - \eta \mathrm{SS})/\gamma\}$, and $\Pr\{\tilde{C} \le B\} \ge 1 - \beta_c$ holds iff $(B - \varphi n - \eta\mathrm{SS})/\gamma \ge q_{1-\beta_c}$, i.e. $\varphi n + \eta\mathrm{SS} + \gamma\, q_{1-\beta_c} \le B$. Because $\gamma\, q_{1-\beta_c} = (q_{1-\beta_c}/N\mu)\,[\varepsilon(v)N\mu D + \varepsilon_L N\mu d(n)]$, this condition is (25). Under normality $q_{1-\beta_c} = N\mu + z_{\beta_c}\sigma_Q$, so $m = 1 + z_{\beta_c}\,\mathrm{CV}_Q$. □

Proof of Corollary 11. The chance constraint multiplies the inbound and last-mile carbon rates by m (Proposition 10); substituting $m\varepsilon_0$ and $m\varepsilon_L$ for ε_0 and ε_L in (13) and (16) gives the two conditions, the latter scaling only the left side and leaving the pooling-adjusted facility threshold $(\varphi + \eta G)/(F + hG)$ on the right. The same substitution in Proposition 7, with n_K, ω, and a_K unchanged because the cost rates are unchanged, gives the joint conditions. From $\sigma_Q^2 = \sigma^2 N[1 + (N-1)\rho]$, $\mathrm{CV}_Q = (\sigma/\mu)\sqrt{[1 + (N-1)\rho]/N}$ is increasing in σ/μ and, for $N > 1$, in ρ, and z_{β_c} is increasing in $1 - \beta_c$, so $m = 1 + z_{\beta_c}\mathrm{CV}_Q$ is increasing in all three. For the joint speed, divide the carbon density condition by m: its facility and lost-pooling terms then scale as m^{-1} and $m^{-4/5}$ and, multiplied by $n^{3/2}$, rise in n with elasticities $\frac{3}{2}$ and at least $\frac{9}{10}$, so that, at an interior n_C, $\mathrm{d}\ln n_C/\mathrm{d}\ln m \le \frac{8}{9}$. Since Ψ has elasticity at most one in n, (22) with $a_K R_{\mathrm{speed}}/m$ in place of $a_K R_{\mathrm{speed}}$ gives $\mathrm{d}\ln v_C/\mathrm{d}\ln m \le -\frac{2}{5} + \frac{1}{5}\cdot\frac{8}{9} < 0$. □